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ADVANTEST CORPORATION

T5300 SERIES

FLASH MEMORY PARALLEL

MEASUREMENT FUNCTION

DESCRIPTION

MANUAL NUMBER 8239605-15

Applicable OSs

ASX/U-50 ASX/U-51

Applicable Systems

<i>T5335</i>	<i>T5377</i>
<i>T5335P</i>	<i>T5376</i>
<i>T5334</i>	<i>T5375</i>
	<i>T5371</i>
	<i>T5336</i>
	<i>T5311</i>

PREFACE

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- (2) The description on failure memory is not applied to the T5336 and T5311 unless otherwise specified.

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

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T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

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1 OVERVIEW

1.1 Outline of This Manual

This manual describes the flash memory parallel measurement function of the T5300 Series Memory Test System and how to generate a program to use this function.

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

1.2 Flash Memory Parallel Measurement Function Block Diagram

1.2 Flash Memory Parallel Measurement Function Block Diagram

The flash memory parallel measurement function comprises the hold, counter/limit, and fail stack registers for each DUT. Figure 1-1 shows the flash memory parallel management function block diagram. (Fail memory is not installed in the T5336 and T5311.)

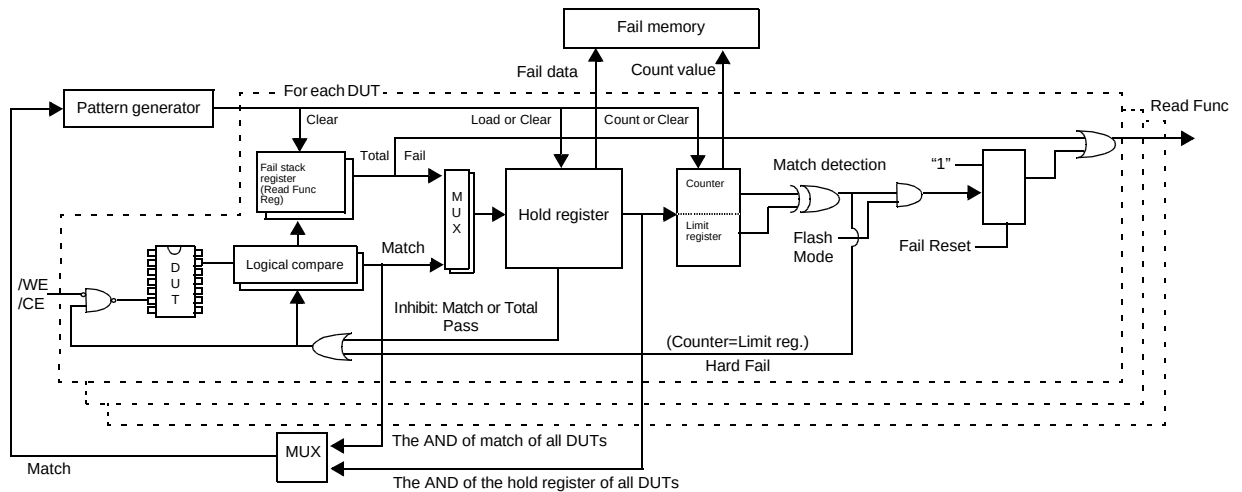


Figure 1-1 Flash Memory Parallel Measurement Function Block Diagram

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

1.3 Overview of Flash Memory Parallel Measurement Function

1.3 Overview of Flash Memory Parallel Measurement Function

This function allows easy and fast parallel measurement of flash memory which has different write/erase characteristics for each device and requires determining pass/fail of device tests by write/erase count and preventing excess write/erase.

This function has the following features:

- (1) Controls prevention of excess write/erase for each device independently by inhibiting in realtime the driver signals such as write enable (/WE) and chip enable (/CE) for rewrite/erase based on the verify output data from the device after write/erase.
- (2) Determines pass/fail in comparison with the limit count for the device test by counting in realtime the write/erase count for each device, based on the verify output data from the device after write/erase.
- (3) Facilitates device fault analysis and error recovery processing by storing the verify output data for each of address and data bits into the file memory when the write count has reached the limit count.
- (4) Allows to store the write count for each address to the fail memory for write tests, facilitating the write count analysis.
- (5) Stacks the verify results for the continuous areas, based on which excessive write/erase is prevented and tests can be performed using the pass/fail decision function by count. This function facilitates the flash memory parallel test of the NAND type.

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

1.4 Overview of Flash Memory Parallel Measurement Software

1.4 Overview of Flash Memory Parallel Measurement Software

The flash memory parallel measurement function can be used with the following versions.

ASX/U-50 REV 6.00 or later

ASX/U-51 REV 3.00 or later

The flash memory parallel measurement statements are compiled by the test plan program compiler.

2 FLASH MEMORY PARALLEL MEASUREMENT STATEMENTS

2.1 Overview

This section explains the statements used to set the mode or data for the flash memory parallel measurement function.

The statements available for the flash memory parallel measurement unit are given below.

- | | | |
|-----|-----------------------|--|
| (1) | SELECT FLASH MODE: | Selects the flash mode.
Sets hold register load or clear conditions.
Sets the flash counter increment or clear conditions.
Sets the type of pins whose output is to be inhibited. |
| (2) | SELECT FLASH DISABLE: | Disables the operation of the flash memory parallel measurement function. |
| (3) | SET FLASH LIMIT: | Sets the write count and erase count limit values for each DUT. |
| (4) | READ FLASH HOLD: | Reads the hold register states for each pin. |
| (5) | READ FLASH COUNTER: | Reads the counter (flash counter) value indicating the write and erase counts for each DUT. |
| (6) | MEAS/START MPAT(5): | Performs the fast erase test. |

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

2.2 SELECT FLASH MODE Statement

2.2 SELECT FLASH MODE Statement

(1) Function

The SELECT FLASH MODE statement is used to validate the flash operation mode and to select the flash mode, set the hold register load and clear conditions, set the flash counter increment and clear conditions, and set the type of pins to be inhibited the driver output.

For information on inhibited driver pins, see Table 3-1.

(2) Format

```
SELECT FLASH [MODE1] H<a, b> C<c, d> [TYPE1]
              [MODE2]
              [MODE3]
              [MODE4]
              [TYPE2]
              [TYPE3]
              [TYPE4]
```

(2-1) MODE1/MODE2/MODE3/MODE4

Sets the flash operation mode.

For more information, refer to Section 3.2.1, "Flash Operation Mode."

(2-2) H<a, b>

a: Sets the hold register load conditions.

b: Sets the hold register clear conditions.

The following data names can be specified:

Data name	a	b
FLAG	☐	Unavailable
R, W, M1, M2, C0-11	☐	☐
FIXL	☐	☐

When omitted, H<FLAG, FIXL> is set by default.

For more information, refer to Section 3.2.2, "Hold Register Operating Conditions (H<a,b>)."

(2-3) C<c,d>

c: Sets the flash counter increment conditions.

d: Sets the flash counter clear conditions.

The following data names can be specified:

Data name	c	d
FLAG	☐	Unavailable
R, W, M1, M2, C0-11	☐	☐
FIXL	☐	☐

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

2.2 SELECT FLASH MODE Statement

When omitted, C<FLAG, FIXL> is set by default.

For more information, refer to Section 3.2.3, "Using Flash Counter (C<c,d>)".

(2-4) TYPE1/TYPE2/TYPE3/TYPE4

Sets the type of pins where driver output is inhibited.

For the T5335, T5335P and T5334: TYPE1 to TYPE4

For the T5377, T5376, T5375, T5371, T5336 and T5311: TYPE1 or TYPE2

When omitted, TYPE1 is set by default.

For more information, refer to Section 3.2.4, "Determining Driver Output Inhibited Pins".

CAUTION!

1. The SELECT FLASH MODE statement is reset when:

The program starts.

CTRL/C is input.

The SELECT FLASH DISABLE statement is executed.

The RESET TESTER statement is executed.

The reset button on the C-BOX is pressed.

2. The SELECT FLASH MODE statement cannot be executed when the following statement or utility program has been used.

- **SELECT APPLICATION MODE BBM statement**
- **IDSELFM (a utility program used to write individual data for each DUT)**
- **IDSELUBM (utility to use universal buffer memory)**

As a result, you need to disable the above statement or utility program by executing the following statement or utility program before you can execute the SELECT FLASH MODE statement.

- **SELECT APPLICATION MODE BBM DISABLE statement**
- **IDSELCLR (a utility program used to write individual data for each DUT)**
- **IDSELCLR (utility to use universal buffer memory)**

2.3 SELECT FLASH DISABLE Statement

(1) Function

The operation of the flash memory parallel measurement function is disabled.

(2) Format

SELECT FLASH DISABLE

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

2.4 SET FLASH LIMIT Statement

2.4 SET FLASH LIMIT Statement

(1) Function

This statement is used to set the write and erase count limit values on each DUT for the flash memory parallel measurement.

(2) Format

```
SET FLASH LIMIT DUTn d1,d2,...,d32
SET FLASH LIMIT DUT(x) d1,d2,...,d32
SET FLASH LIMIT ALL d
```

- DUTn: : Specifies the DUT number to start the setting.
n must be an integer-type constant 1-128.
- DUT(x): : This format is used when specifying the starting DUT number
with a variable.
x must be an integer-type variable 1-128.
- ALL: : ALL used instead of DUTn or DUT(x) specifies all DUTs.
- $\left[\begin{matrix} d1,d2,...,d32 \\ d \end{matrix} \right]$: Specifies the limit value of the write or erase count.
Specifies an integer or logical constant, integer or logical
variable, non-subscripted array variable or MASK.
When MASK is specified, logical comparison using the driver
mask is inhibited.
- (a) For the T5335, T5335P, T5334, T5371, T5336 and T5311
Specify 0 to 4095 (#FFF), 36863 (#8FFF) or MASK.
MASK has the same effect as the 36863 (#8FFF) setting.
- (b) For the T5376 and T5375
Specify 0 to 65535 (#FFFF) or MASK.
- (c) For the T5377
Specify 0 to 262143(#3FFFF) or MASK.

(3) Operation description

The SET FLASH LIMIT statement is used to set the limit value of the write and erase count in the limit register.

The following methods can be used to set the limit register:

- Setting the limit value common to all DUTs (see Example 1).
- Setting the limit value for each DUT (see Example 2).
- Setting MASK (see Example 3).

When MASK is specified, driver output and logical comparisons are inhibited.

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

2.4 SET FLASH LIMIT Statement

(4) Usage example

Example 1: When a limit value of 25 is set for all DUTs

SET FLASH LIMIT ALL 25

Example 2: When a limit value is set for each DUT.

SET FLASH LIMIT DUT1 25, 24, 23, 22, 21, 20, 19, 18

Example 3: When MASK is set for DUT2, DUT4 and DUT6.

SET FLASH LIMIT DUT2 MASK, 10, MASK, 20, MASK

CAUTION!

1. In the SET FLASH LIMIT statement, MASK cannot be used as a variable name.
2. This statement should not be set for DUT not to be tested. (The DUT to be set is SDUT.)
For more information on SDUT, refer to the following sections:
The section "READ mDUT Statement" in the ATL-50 Test Plan Program Reference Manual
VOL7 System Interface
The section 7.41.1, "READ xDUT" in Chapter 7, "System Interface," of "ATL-51 Test Plan Program Reference Manual."
3. If the SET FLASH LIMIT statement is omitted, the following limit value is set by default.
For the T5335, T5335P, T5334, T5371, T5336 and T5311: 4095 (#FFF)
For the T5376 and T5375: 65535 (#FFFF)
For the T5377: 262143(#3FFFF)
4. If a value exceeding the maximum DUT number is specified for DUTn or DUT(x), a RUNTIME ERROR (3111) occurs.

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

2.5 READ FLASH HOLD Statement

2.5 READ FLASH HOLD Statement**(1) Function**

This statement reads the hold register data for each pin.

This statement is used for information management using the result of the verify test, which is performed after flash memory write and erase operations have been performed.

(2) Format

```
READ FLASH HOLD (pin name) TO v1,v2,...,v32
READ FLASH HOLD (block number) TO v1,v2,...,v32
```

pin name: Specifies a pin number using a constant, a variable or a PINLIST name. For more information, refer to the following manuals:
 ATL-50 Test Plan Program Reference Manual VOL1 PIN SPECIFICATIONS
 ATL-51 Test Plan Program Reference Manual Chapter 1 Pin Conditions Statements
 (Example 1) When specifying PC33 through PC36 using constants.
 READ FLASH HOLD (PC 33-36) TO VBUFF
 (Example 2) When specifying PC33 through PC36 using variables.
 DIM A(4)=33, 34, 35, 36
 READ FLASH HOLD (PC (A)) TO VBUFF
 (Example 3) When specifying PC33 through PC36 using PINLIST names.
 PINLIST DOUT =PC 33-36
 READ FLASH HOLD (DOUT) TO VBUFF

Block number: A pin block number (beginning with 0) is assigned to each of 32 pin channels. The block numbers of comparator pins are as follows:

Block number	Channels
1	33 to 50
3	97 to 114

v1,v2,...,v32: Specifies a variable to store the read data. For the variable, specify a 32-bit logical variable or non-subscripted array variable. Data stored in each bit is explained below.

Data	Driver state	Logical comparison
"1"	Masked	Inhibited
"0"	Not masked	Not inhibited

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

2.5 READ FLASH HOLD Statement

(3) Operation description

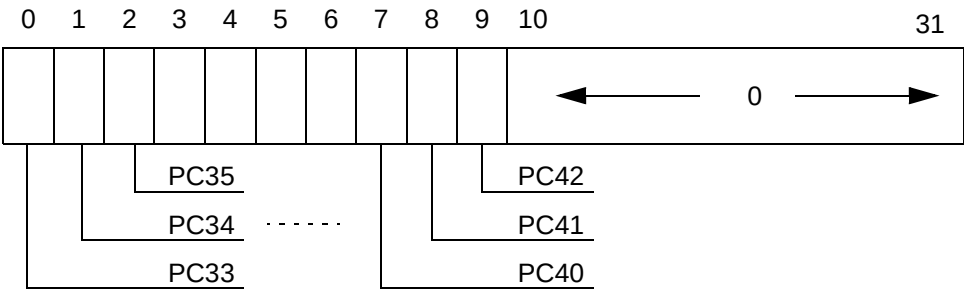
The READ FLASH HOLD statement allows to read the information of each I/O pin from the hold register when the pattern generator is stopped or in wait state. The data read depends on the flash mode settings.

Operation mode	"0"	"1"
MODE1 MODE2	UNMATCH of match detecting result	MATCH of match detecting result
MODE3 MODE4	Total fail state of fail stack register	Pass state of fail stack register

(4) Data storage format

(a) When specifying pins (socket conversion)

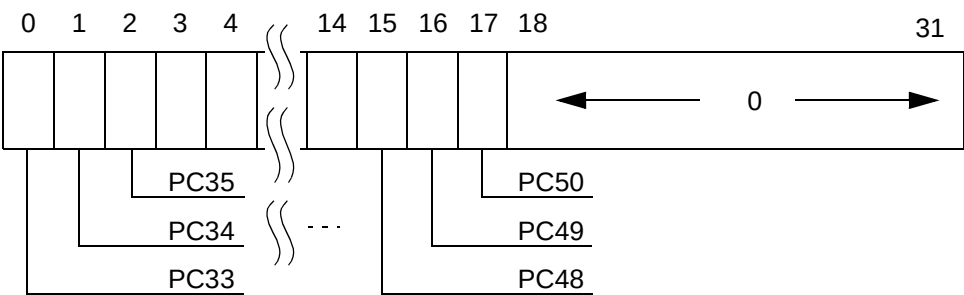
When PC33-42



When data cannot be stored in one variable, it is stored in subsequent variable(s). For the bits related to the pins inhibiting logical compare with driver MASK, "1"s are set.

(b) When specifying block number (no socket conversion)

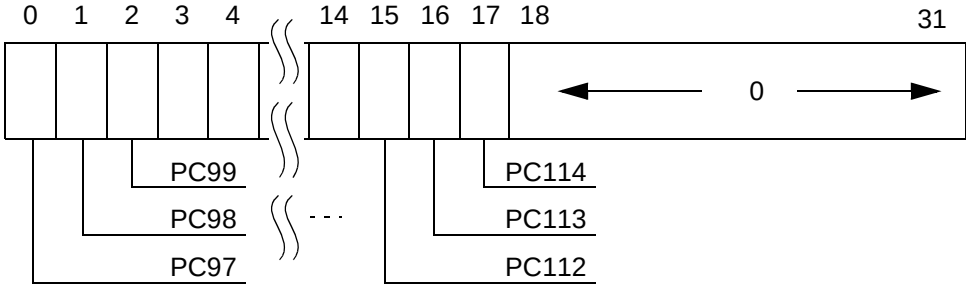
When the block number is "1"



T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

2.5 READ FLASH HOLD Statement

When the block number is "3"



(5) Relationships between DUT numbers and storing variables for parallel measurement

Example 1: Parallel measurement of four DUTs

DIM[2] X(4)

⌋

READ FLASH HOLD (PC33-40) TO X

This causes the following DUT numbers to be stored in the variables:

Variable	X(1)	X(2)	X(3)	X(4)
DUT number	DUT1	DUT2	DUT3	DUT4

Example 2: Parallel measurement of eight DUTs of two STNs

DIM[2] X(8)

⌋

READ FLASH HOLD (PC33-40) TO X

This causes the following DUT numbers to be stored in the variables:

Variable	X(1)	X(2)	X(3)	X(4)
DUT number	DUT1	DUT2	DUT3	DUT4

; 1'st STN

Variable	X(5)	X(6)	X(7)	X(8)
DUT number	DUT5	DUT6	DUT7	DUT8

; 2'nd STN

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

2.5 READ FLASH HOLD Statement

CAUTION!

1. *Measurement can be started with any DUT by using a SET RDUT statement which specifies the first reading DUT for parallel measurement. Normally, the measurement starts with DUT1.*
2. *For DUT not under test, the data does not exist and the corresponding storing variables remain unchanged. (The DUT to be read is SDUT.)*
3. *For pin and block number specifications, pins other than comparator pins can also be specified. In this case, "0"s are stored as the result.*
4. *The hold register is cleared by each MEAS/START MPAT statement, and then the test is carried out.
In START MPAT(1) (Continue mode), the hold register is not cleared.*

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

2.6 READ FLASH COUNTER Statement

2.6 READ FLASH COUNTER Statement

(1) Function

This statement reads the flash counter (counting the write and erase counts for the flash memory parallel measurement) for each DUT.

(2) Format

```
READ FLASH COUNTER TO v1,v2, ...,v32
```

v1,v2,...,v32: Specify the storing variables with integer-type or logical-type variables. An all-specify type variable DIM is also available.

(3) Operation description

The READ FLASH COUNTER statement is used to read the flash counter value (hereafter, referred to as the counter value) of each DUT set when the pattern generator stops operating or is in a wait state.

This statement is used, for example, when you want to use the erase count for each DUT (which is obtained from the previously executed erase test) for controlling the limit value of the erase count for the next erase test in the test plan program

The counter value ranges from 0 to the flash limit value (hereafter, referred to as limit value).

When the counter value matches the limit value, the following bits are set to 1:

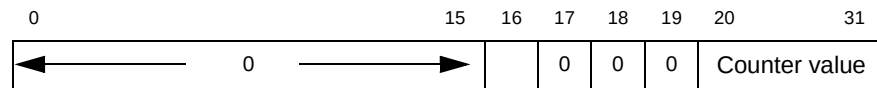
For the T5335, T5335P, T5334, T5371, T5336 and T5311 : Bit 16

For the T5377, T5376 and T5375 : Bit 0

(4) Data storage format

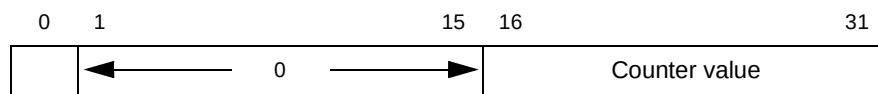
The counter value is stored in a variable in the following format.

For the T5335, T5335P, T5334, T5371, T5336 and T5311



Set to "1" when the counter value matches the limit value.

For the T5376 and T5375

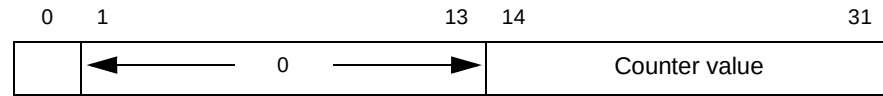


Set to "1" when the counter value matches the limit value.

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

2.6 READ FLASH COUNTER Statement

For the T5377



Set to "1" when the counter value matches the limit value.

(5) Example

```
LOGI [2] DAT (16)
READ FLASH COUNTER TO DAT
```

CAUTION!

1. *Measurement can be started with any DUT by using a SET RDUT statement which specifies the first reading DUT for parallel measurement. Normally, the measurement starts with DUT1.*
2. *Variables corresponding to the devices which were not tested are not changed because no new data was assigned to them. (This command assigns the number of times the memory devices were erased and reprogrammed to the corresponding variable.)*
3. *The flash counter is cleared to 0 by each MEAS/START MPAT statement, and then the test is carried out. In START MPAT(1) (Continue mode), the flash counter is not cleared.*

2.7 MEAS/START MPAT(5) Statement

(1) Function

This statement can execute the fast erase test easily by using the WAIT command of the pattern program together.

(2) Format

MEAS	MPAT (5)	pattern file name
START	MPAT (5)	pattern file name

(3) Operation description

The pattern program is interrupted temporarily by the WAIT instruction of the pattern program. In the mean time, the following processes are executed by the tester driver controller.

- The first failed X address is set to the XH register.
- The first failed Y address is set to the YH register.
- The verify length (the number of loops) from the first failed address is set to the IDX8 register.
- The value of the address having WAIT instruction +1 is set to the PC register.
- The fail address is cleared.
- Continuous start (Restart of the pattern program.)

Refer to “4.4 Fast Erase Test on Flash Memory Using MEAS MPAT(5)” for details.

CAUTION!

1. *It takes 80 μ s to process the WAIT state when the MEAS/START MPAT(5) statement is executed.*
2. *When MEAS/START MPAT(5) is used, the SELECT FLASH MODE2 statement or SELECT FLASH MODE4 statement must be execute before that.*
3. *To conform the number of loops to STI8 n written in the pattern program, write IDX8=n in the REGISTER instruction section of the pattern program or write REG MPAT IDX8=n in the test plan program.*
4. *The MEAS/START MPAT(5) statement allows the first failed address to be transferred to the XH and YH registers when the test system is restarted from a wait state. In the cycle when the ALPG is restarted from a WAIT state, it is necessary to write arithmetic address instructions such as XB<XH YB<YH in the pattern program and to load addresses to be verified. For more information on programming, refer to Section 4.4.3 “Sample Program 1 for the Fast Erase Test Method (Chip Erase).”*

3 OVERVIEW PROGRAMMING

3.1 Overview

This section describes how to program flash memory tests using the flash memory parallel measurement functions and their operations.

3.2 Programming and Operations of SELECT FLASH MODE Statement

The SELECT FLASH MODE statement is used to set the following conditions to use the flash memory parallel measurement functions:

- (a) Flash operation mode (MODE1 to MODE4, DISABLE)
- (b) Hold register operating conditions (H<a,b>)
- (c) Flash counter operating conditions (C<c,d>)
- (d) Selection of a pin which inhibits driver output to prevent excessive write/erase (TYPE1-4)

This section explains the conditions and operations listed above.

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

3.2 Programming and Operations of SELECT FLASH MODE Statement

3.2.1 Flash Operation Mode

The flash memory parallel measurement functions are controlled by five operation modes; MODE1, MODE2, MODE3, MODE4, DISABLE.

(1) MODE1

In this mode, the flash memory parallel measurement functions are controlled using the match detecting function information which allows to detect matching between the device output data and expected data (EXP).

The match detecting function is executed for the I/O pins (P33-50, and P97-114) set as MATCH in the pin setting. The compare result is assumed as MATCH when the expected values for the VOH and VOL of the device output data match the result values, and as UNMATCH when they do not match. The decision timing is STRB1 when using edge strobes and WSTRB1 when using multi-window strobes. For pins not specified MATCH, MATCH is always assumed.

The verify result of flash memory write/erase is determined using the match detecting function and the test is carried out while deciding whether to perform write/erase again.

This mode is mainly used for write/erase tests on NOR-type flash memory.

(2) MODE2

This mode allows to logically compare the device output data in a series of areas by pass/fail, stacks the fail information in the fail stack register for each I/O pin, and controls the flash memory parallel measurement functions using this stacked fail information.

This mode is mainly used to test NAND-type flash memory while deciding to perform write/erase again in block units.

(3) MODE3

An operation mode adding the following condition to MODE1.

When the flash counter value matches the limit value, the flash counter clear function is disabled.

(4) MODE4

An operation mode adding the following condition to MODE2.

When the flash counter value matches the limit value, the flash counter clear function is disabled.

(5) DISABLE

This mode inhibits the flash memory parallel measurement functions.

CAUTION!

1. ***When MODE1 to MODE4 has been set, data is loaded to the hold register to clear the fail stack register. (Since the content of the fail stack register is also read by READ FUNC, data load to the hold register changes the pass/fail result generated by READ FUNC.)***
2. ***When the flash counter value matches the limit value in MODE3 or MODE4, the flash counter clear function is disabled. However, changing the limit value to a value different from the flash counter value while the pattern program is running enables the clear function and allows the flash counter to be cleared.***

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

3.2 Programming and Operations of SELECT FLASH MODE Statement

3.2.2 Hold Register Operating Conditions (H<a,b>)

In MODE1 or MODE3, the hold register loads the match detecting information (MATCH/UNMATCH) for each I/O pin under the conditions specified in “a” of H<a,b>. In MODE2 or MODE4, it loads the fail stack register information (PASS/FAIL) under the same conditions. The logical product of the hold register information for individual I/O pins is calculated in the device I/O bit width to control whether to inhibit the driver output and logical compare for each I/O pin.

Inhibiting of the driver output and logical compare is cleared by “b” of H<a,b> which specifies the hold register information.

In the flash operation mode, the match detection is performed for all DUTs with reference to the hold register information (as shown in Figure 1-1).

Because the hold register contains the cycles in which match flag instructions are written, match detection cannot be performed if the hold register is cleared.

The Table 3-1 below gives the hold register input information, load conditions, clear conditions, clear states, and the conditions to inhibit the driver output and logical compare.

Table 3-1 Relationships between Operation Mode and Hold Register

Operation mode	Hold register				Conditions to inhibit driver output and logical compare
	Input information	Load conditions	Clear conditions	Clear states	
MODE1 MODE3	Match detecting information	Conditions of “a” specified in H<a,b>	Conditions of “b” specified in H<a,b>	UNMATCH state	All bits are MATCH
MODE2 MODE4	Fail information in fail stack register			FAIL state	All bits are PASS

In H<a,b>, possible “a” and “b” values are as follows:

Table 3-2 H<a,b> Specification and the Execution Cycle in Pattern Program

Data name	“a”	“b”	Execution cycle
FLAG	○	Unallowable	Cycles for which match flag sense instructions are described in the pattern program.
R, W, M1, M2 C0-11	○	○	Cycles for which the MUT command given in the left is described.
FIXL	○	○	Inhibits operations in every cycle.

Inhibit or clear of the driver output and logical compare by the hold register are effective from the cycle succeeding the one specifying the load and clear conditions.

For the cycle specifying the load and clear conditions for the hold register, the rate which satisfy the following conditions must be specified:

For the T5377, T5376 and T5375

$$\text{RATE} \geq \text{STRB1 setting} + 230 \text{ ns}$$

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

3.2 Programming and Operations of SELECT FLASH MODE Statement

For the T5371

$\text{RATE} \geq \text{STRB1 setting} + 170 \text{ ns}$ and the clock set value for the pins where driver output is inhibited $\geq 10 \text{ ns}$

For the T5335, T5335P, T5334, T5336 and T5311

$\text{RATE} \geq \text{STRB1 setting} + 150 \text{ ns}$

When a multi-window strobe is specified, the STRB1 set time becomes the WSTRB1 trailing set time.

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

3.2 Programming and Operations of SELECT FLASH MODE Statement

3.2.3 Using Flash Counter (C<c,d>)

The flash counter operates when the flash operation mode (MODE1 to MODE4) is set.

The flash counter increments for each device when the conditions of “c” in C<c,d> and the hold device information of each device unmatched in MODE1 or MODE3, or when they fail in MODE2 or MODE4.

In the sequence which allows rewrite/erase when the verify result of a write/erase test is unmatched, the write/erase operations can be counted by checking the verify result.

The flash counter is cleared by the conditions of “d” in C<c,d>. It can count the writes at each address for a write test, or count erases of the entire chip or each block for an erase test. The count values are compared with the limit values of write/erase counts set in the limit register to decide pass/fail by the counts. When the count value and the limit value match, it is determined as a fail. In this case, the driver output and logical compare are inhibited in order to prevent excessive write/erase on the failed device.

The limit value is set with a SET FLASH LIMIT statement.

The count value in the flash counter becomes effective from the cycle succeeding the one specifying the increment and clear conditions.

In C<c,d>, possible “c” and “d” values are as follows:

Table 3-3 C<a,b> Specification and the Execution Cycle in Pattern Program

Data name	“a”	“b”	Execution cycle
FLAG	○	Unallowable	Cycles for which match flag sense instructions (FLGL1, FLGLI1, HFLG-LI, etc.) are described in the pattern program.
R, W, M1, M2 C0-11	○	○	Cycles for which the MUT command given in the left is described.
FIXL	○	○	Inhibits operations in every cycle.

For the cycle specifying the increment and clear conditions for the flash counter, the rate which satisfy the following conditions must be specified:

For the T5377, T5376 and T5375

$$\text{RATE} \geq \text{STRB1 setting} + 230 \text{ ns}$$

For the T5371

$$\text{RATE} \geq \text{STRB1 setting} + 170 \text{ ns and the clock set value for the pins where driver output is inhibited} \geq 10 \text{ ns}$$

For the T5335, T5335P, T5334, T5336 and T5311

$$\text{RATE} \geq \text{STRB1 setting} + 150 \text{ ns}$$

When a multi-window strobe is specified, the STRB1 set time becomes the WSTRB1 trailing set time.

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

3.2 Programming and Operations of SELECT FLASH MODE Statement

3.2.4 Determining Driver Output Inhibited Pins

Inhibited pins can be determined from the settings of the socket program and the SELECT FLASH MODE statement.

- ### (1) Socket program settings

For the socket program, set BIT and PIN MODE, and select driver pins and comparator pins.

- (2) SELECT FLASH MODE statement setting

For the SELECT FLASH MODE statement, select TYPE.

Changing the TYPE setting in the test plan program can change the target pin.

Example: When BIT=36 and PIN MODE=A are set

According to the TYPE, BIT and PIN MODE settings shown in

, the following pins can be specified.

For the T5335, T5335P and T5334:

	29A1 for TYPE1
	30A1 for TYPE2

For the T5377, T5376, T5375, T5371, T5336 and T5311:
26A1 for TYPE1
29A1 for TYPE2

- Socket program

```

SOCKET SOC1
└
  BIT=36
  PIN MODE=A
  └

```

- Test plan program

```

PRO PR01 SOC1
└
  SELECT FLASH MODE1 H<FLAG, FIXL> C<FLAG, FIXL> TYPE1
  MEAS MPAT
└
  SELECT FLASH MODE1 H<FLAG, FIXL> C<FLAG, FIXL> TYPE2
  MEAS MPAT
└

```

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

3.2 Programming and Operations of SELECT FLASH MODE Statement

(3) Specifying inhibited driver pins.

Up to four driver pins can be allocated for each DUT when using the T5335, T5335P or T5334.

Up to two driver pins can be allocated for each DUT when using the T5377, T5376, T5375, T5371, T5336 or T5311.

UP to two pins can be specified as the pins where driver output is inhibited for one DUT at a time.

Driver pins that can be inhibited are fixed as shown in Table 3-4 below.

Table 3-4 Driver Pins That Can Be Inhibited

- For the T5335, T5335P and T5334

(*1)		Socket program setting					
TYPE	PIN MODE (*2)	Pin configur ation	Comparator pin group				BIT
			33-40, 49	41-48, 50	97-104, 113	105-112, 114	
1	A	×1	29A1				32, 36
		×2	29A1				32, 36
			29A1		29A2		16, 18
	B	×2	29A1	30A1	29A2	30A2	8, 9
		×4	29A1	29A3	29A2	29A4	8, 9
2	A	×1	30A1				32, 36
		×2	30A1				32, 36
			30A1		30A2		16, 18
	B	×2	93A1	94A1	93A2	94A2	8, 9
		×4	30A1	30A3	30A2	30A4	8, 9
3	A	×1	93A1				32, 36
		×2	93A1				32, 36
			93A1		93A2		16, 18
4	A	×1	94A1				32, 36
		×2	94A1				32, 36
			94A1		94A2		16, 18

*1 TYPE is set in the SELECT FLASH MODE statement.

*2 Child B through H have the same pin configuration as Child A.

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

3.2 Programming and Operations of SELECT FLASH MODE Statement

- For the T5377, T5376 and T5375 (when using the socket program which sets PD1 through PD30A1 for one DUT), T5371, T5336 and T5311

(*1)		Socket program setting				
TYPE	PIN MODE	Comparator pin group				BIT
		33-40A1, 49A1	41-48A1, 50A1	97-104A1, 113A1	105-112A1, 114A1	
1	A	26A1 (*2)				32, 36
		26A1		27A2		16, 18
	B	26A1	29A1	27A2	30A2	8, 9
2	A	29A1 (*3)				32, 36
		29A1		30A2		16, 18

*1 *TIPE is set in the SELECT FLASH MODE statement.*

*2 *Pin 27 cannot be used with the T5336 and T5311.*

*3 *Pin 30 cannot be used with the T5336 and T5311.*

Note:

- When A1 and A2 have the same pin number, they cannot be used at the same time.
If driver output is inhibited for 26A1, 26A2 cannot be used.
If driver output is inhibited for 27A2, 27A1 cannot be used.
If driver output is inhibited for 29A1, 29A2 cannot be used.
If driver output is inhibited for 30A2, 30A1 cannot be used.
- For the T5377, T5376 and T5375
Child A3, A5 and A7 have the same pin configuration as Child A1.
Child A4, A6 and A8 have the same pin configuration as Child A2.

For the T5371, T5336 and F5311
Child A3 has the same pin configuration as Child A1.
Child A4 has the same pin configuration as Child A2.
- Child B through H have the same pin configuration as Child A.
- When the socket program is not used, TYPE1, PIN MODE=A and BIT=36 are specified.

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

3.2 Programming and Operations of SELECT FLASH MODE Statement

- For the T5377, T5376 and T5375 (when using the socket program which sets PD1through 30A1 for two or more DUTs)

(*1)	Socket program setting					
TYPE	PIN MODE	Comparator pin group				BIT
		33-40A1, 49A1	41-48A1, 50A1	97-104A1, 113A1	105-112A1, 114A1	
1	A	26A1				32, 36
		26A1		26A5		16, 18
	B	26A1	27A2	26A5	27A6	8, 9
2	A	29A1				32, 36
		29A1		29A5		16, 18
	B	29A1	30A2	29A5	30A6	8, 9

*1 TYPE is set in the SELECT FLASH MODE statement.

Note:

- When A1 and A2 have the same pin number, they cannot be used at the same time.
 If driver output is inhibited for 26A1, 26A2 cannot be used.
 If driver output is inhibited for 27A2, 27A1 cannot be used.
 If driver output is inhibited for 29A1, 29A2 cannot be used.
 If driver output is inhibited for 30A2, 30A1 cannot be used.

 If driver output is inhibited for 26A5, 26A6 cannot be used.
 If driver output is inhibited for 27A6, 27A5 cannot be used.
 If driver output is inhibited for 29A5, 29A6 cannot be used.
 If driver output is inhibited for 30A6, 30A5 cannot be used.
- Child A3 has the same pin configuration as Child A1.
 Child A4 has the same pin configuration as Child A2.
 Child A7 has the same pin configuration as Child A5.
 Child A8 has the same pin configuration as Child A6.
- Child B through D have the same pin configuration as Child A.
- When the socket program is not used, TYPE1, PIN MODE=A, and BIT=36 are specified.

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

3.2 Programming and Operations of SELECT FLASH MODE Statement

(4) Usage example

To inhibit driver signals of Write Enable (/WE) or Chip Enable (/CE)

Controlling the /WE and /CE signal outputs can prevent the excessive write or excessive erase of the flash memory.

The following example explains how to set both /WE and /CE for the driver pin where the output is inhibited for one DUT.

When the following socket program is used and TYPE1 is specified in the SELECT FLASH MODE statement, the /WE pin is set to the pin where driver output is inhibited. When TYPE2 is specified, the /CE pin is set to the pin where driver output is inhibited.

- Socket program (for the T5335, T5335P and T5334)

```

┌
DUT# = ..., PD29,      PD30,      P33-40
DUT1 = ..., PD29A1,   PD30A1,    P33-40A
DUT2 = ..., PD29A3,   PD30A3,    P41-48A
DUT3 = ..., PD29A2,   PD30A2,    P97-104A
DUT4 = ..., PD29A4,   PD30A4,    P105-112A
└

```

Note: *BIT=8, PIN MODE=B, Pin configuration x4, /WE pin=PD29, /CE pin=PD30*

- Socket program (for the T5377, T5376, T5375, T5371, T5336 and T5311)

```

┌
DUT# = ..., PD26,      PD29,      P33-40
DUT1 = ..., PD26A1,   PD29A1,    P33-40A1
DUT2 = ..., PD26A3,   PD29A3,    P33-40A3
DUT3 = ..., PD27A2,   PD30A2,    P97-104A1
DUT4 = ..., PD27A4,   PD30A4,    P97-104A3
└

```

Note: *BIT=8, PIN MODE=A, /WE pin=PD26, /CE pin=PD29*

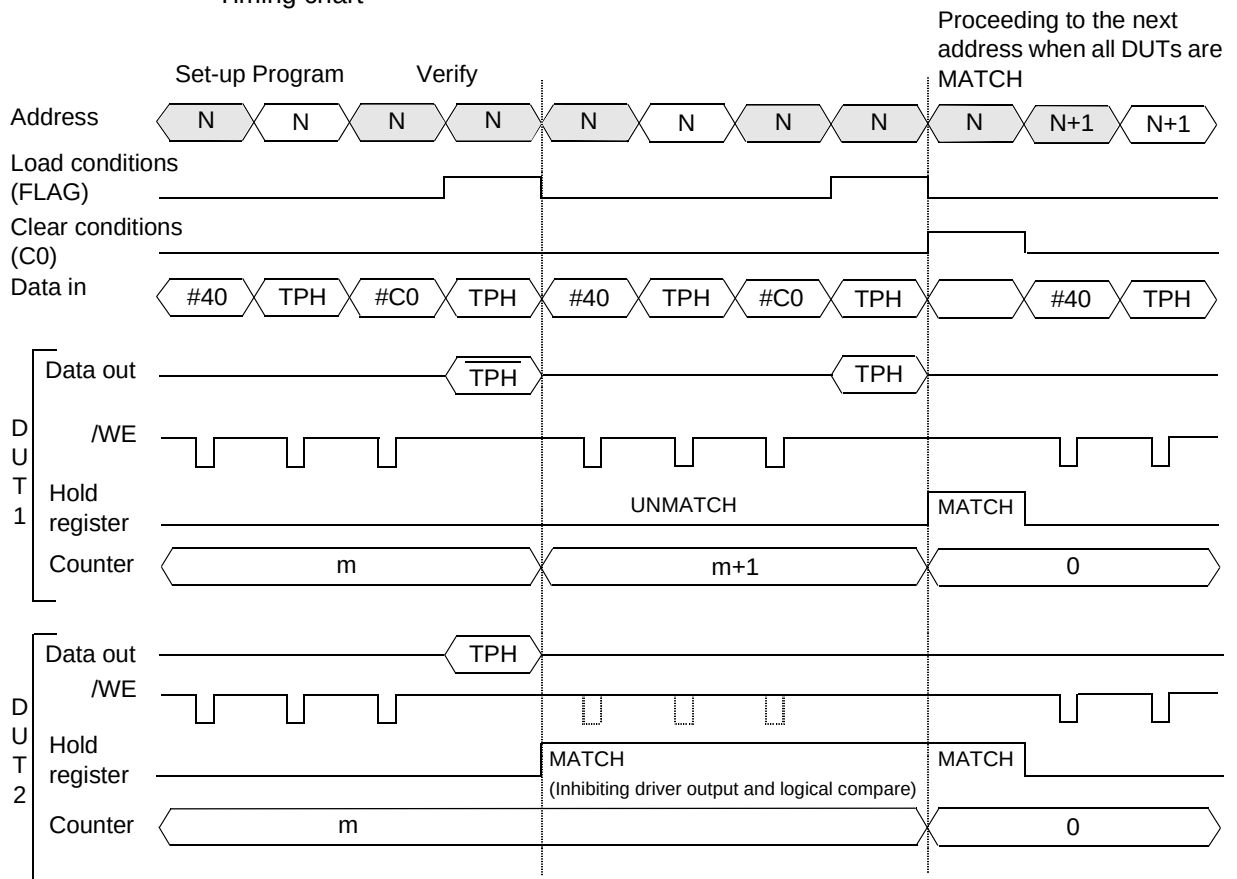
T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

3.2 Programming and Operations of SELECT FLASH MODE Statement

- Pattern program

	NOP		XB<0 YB<0		
	NOP		TP<#40		TPH: PROGRAM DATA
ST0:	NOP	W	TP<TPH		#40: PROGRAM SET-UP COMMAND
	NOP	W	TP<#C0		#C0: PROGRAM VERIFY COMMAND
	NOP	W	TP<TPH		
	<u>FLGLI1</u>	<u>ST0</u>	TP<#40		
	JNI1	ST0	<u>C0</u>	XB<XB+1 YB<YB+1 ^ BX	

- Timing chart



Note: The output voltage of a driver pin where driver output is inhibited is fixed to the V_{IH} level set in the pin statement.

Specify the negative logic waveform type for the pins where driver output is inhibited.

(Example) When specifying the negative logic NRZ waveform for the /WE pin
WE=.... /NRZ.....

The driver output and logical comparison remain inhibited until the next functional test is executed. To reset the setting, specify the SELECT FLASH DISABLE statement.

3.3 Pattern Program

If "FLAG" is specified as a data name in the SELECT FLASH MODE statement, the FLAG execution cycle becomes the cycle when a match flag sense instruction is specified in the pattern program.

The following match flag sense instructions are available.

For more information, refer to the following manuals:

Section 2.3, "Match Flag Sense Instruction of the Pattern Sequence Control Section," in the ATL-50 Pattern Program Reference Manual

Section 2.3, "Match Flag Sense Instruction of the Pattern Sequence Control Section," in the ATL-51 Pattern Program Reference Manual

FLGW1

FLGJ1

FLGS1

FLGL1

FLGLI1

FLGLIZ (Cannot be executed with the T5336 and T5311)

HFLGL1 (Cannot be executed with the T5377, T5376, T5375 and T5371)

FJN18 (Cannot be executed with the T5377, T5376, T5375 and T5371)

Set RATE as follows in the cycle when the match flag sense instruction is specified.

Table 3-5 RATE Condition for the Cycle for Which the Match Flag Sense Instruction is Specified

Test system	RATE condition
T5377	RATE $\geq$ STRB1 value +1.0 μ s (Note1)
T5376, T5375	RATE $\geq$ STRB1 value +1.0 μ s (Note2)
T5371	RATE $\geq$ STRB1 value +1.0 μ s
T5336, T5311	RATE $\geq$ STRB1 value +1.4 μ s

Note1: When the mask pattern is generated by the AFM.

When the AFM RATE statement specifies HS or MS.

RATE $\geq$ STRB1 value + 1.6 μ s

When the AFM RATE statement specifies LS.

RATE $\geq$ STRB1 value + 2.0 μ s

When the mask pattern is not generated by the AFM.

When the AFM RATE statement specifies LS.

RATE $\geq$ STRB1 value + 1.4 μ s

Note2: When the AFM RATE statement specifies LS.

RATE $\geq$ STRB1 value + 1.4 μ s

3.4 Socket Program

According to Table 3-1, set BIT and PIN MODE and select comparator pins and driver pins.

For the BIT, specify 8, 9, 16, 18, 32 or 36.

If the above socket program settings are omitted, BIT=36 and PIN MODE=A are assumed.

3.5 Using Address Failure Memory (AFM)

Using AFM enables the flash counter value and hold register data for a desired cycle to be stored in the AFM during pattern program execution.

In a test plan program, specify the AFM ACTION statement and READ AFM statement after the SELECT FLASH MODE statement is specified.

(The AFM cannot be used for the T5336/T5311.)

```
PRO FLPR000X

SELECT FLASH MODE statement
└
AFM ACTION statement
└
```

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

3.5 Using Address Failure Memory (AFM)

3.5.1 Flash Counter Value

The flash counter value can be read by specifying AF mode in the AFM ACTION statement and the READ AFM statement.

The size specified by the AFM BITSIZE statement is also used.

The AF mode consists of AF(sg), AFSBn(sg) and AFSB(x)(sg).

For the “sg”, RD, WT, M1, M2 or C0 to C11 can be specified for the cycle when the counter value is to be stored.

Example: When 8-bit data is stored in the cycle when C1 is specified

```
AFM BITSIZE = 8
AFM ACTION  = AF(C1)>FMX, .....
      ∫
MEAS MPAT .....
      ∫
READ AFM AF .....
```

The maximum size of data to be stored is as follows.

T5335, T5335P, T5334 and T5371: 8 bits

T5377, T5376 and T5375: 9 bits

For more information on the format used to read AFM data, refer to the following sections of the manual.

Section 2.5, “READ AFM Memory Statement,” in the ATL-50 Test Plan Program Reference Manual VOL10 Failure Analysis Memory

Section 10.23.2, “Data Formats of the Flash Counter,” in Chapter 10, “Failure Analysis Memory,” of “ATL-51 Test Plan Program Reference Manual.”

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

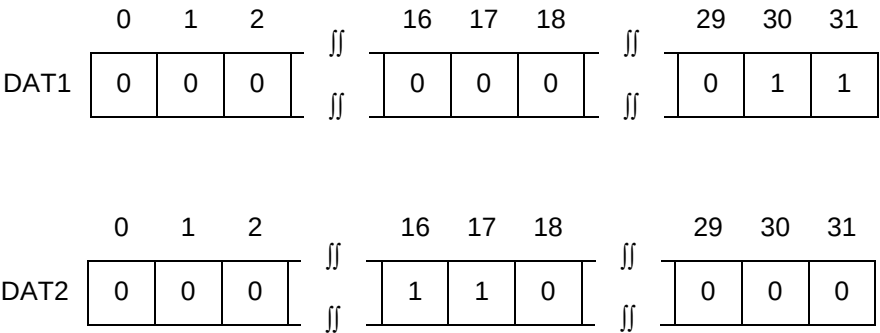
3.5 Using Address Failure Memory (AFM)

Note:

1. When the flash counter value has been stored, clear the AFM contents for each test.
2. The READ FLASH COUNTER statement and READ AFM statement use different formats.
 The READ FLASH COUNTER statement is used to read the flash counter values.
 The READ AFM statement is used to read out data to the specified bit positions.

Example: When the flash counter value is #3

```
LOGI[2] DAT1,DAT2
READ FLASH COUNTER TO DAT1 ;; DAT1=#3
READ AFM AF .....TO DAT2 ;; DAT2=#C000
```



T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

3.5 Using Address Failure Memory (AFM)

3.5.2 Hold Register Data

The hold register data can be read by specifying BF mode in the AFM ACTION and READ AFM statements.

For the size of bits to be read, the size specified by the AFM BITSIZE statement is also used.

The BF mode consists of BF(sg), BFSBn(sg) and FSB(x)(sg).

For the "sg", RD, WT, M1, M2 or C0 to C11 can be specified for the cycle when the hold register data is stored.

Example: When 8-bit data is stored in the cycle when C2 is specified.

```
AFM BITSIZE = 8
AFM ACTION  = . . . . ., BF(C2)>FMX
└
MEAS MPAT . . . . .
└
READ AFM BF . . . . .
```

Channels that can be used to read data are listed below.

Channel	READ FLASH HOLD statement	READ AFM statement
33 to 50	READ FLASH HOLD(1) TO V	READ AFM BF ... BLK1 TO V
97 to 114	READ FLASH HOLD(3) TO V	READ AFM BF ... BLK2 TO V

For more information on the format used to read AFM data, refer to the following sections:

Section 2.5, "READ AFM Memory Statement," in the ATL-50 Test Plan Program Reference Manual VOL10 Failure Analysis Memory

Section 10.20.2, "Data format," in Chapter 10, "Failure Analysis Memory," of "ATL-51 Test Plan Program Reference Manual."

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

3.5 Using Address Failure Memory (AFM)

Note:

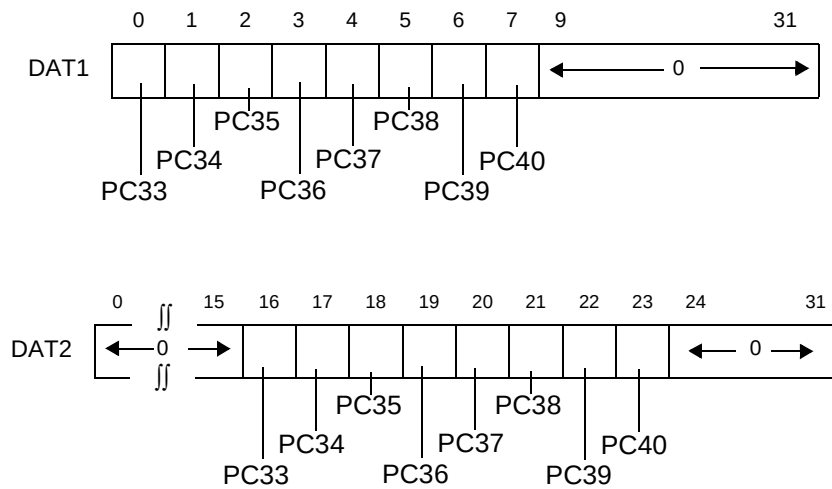
The READ FLASH HOLD statement and READ AFM statement use different formats for reading data.

Example: When 8-bit data (channels 33 to 40) of the hold register is read

LOGI[2] DAT1,DAT2

READ FLASH HOLD(1) TO DAT1

READ AFM BF DUT1 BLK1 TO DAT2



4 FLASH MEMORY PARALLEL MEASUREMENT FUNCTION PROGRAMMING EXAMPLE

This section gives the following examples using the flash memory parallel measurement functions and describes about these examples:

For test plan program and socket program examples, note that some hardware pin number configurations and other configurations differ depending on the test system but the pattern program examples do not differ in this way.

- (1) NOR-type flash memory write test example
- (2) NOR-type flash memory erase test example
- (3) Program example for verifying flash memory in block units
- (4) Fast erase test on flash memory using MEAS MPAT(5)

4.1 NOR-type Flash Memory Write Test Example

In a write test, the written data is verified. If it is not correct, rewriting must be done by incrementing the write count. Therefore, the tester match detecting function is used for testing.

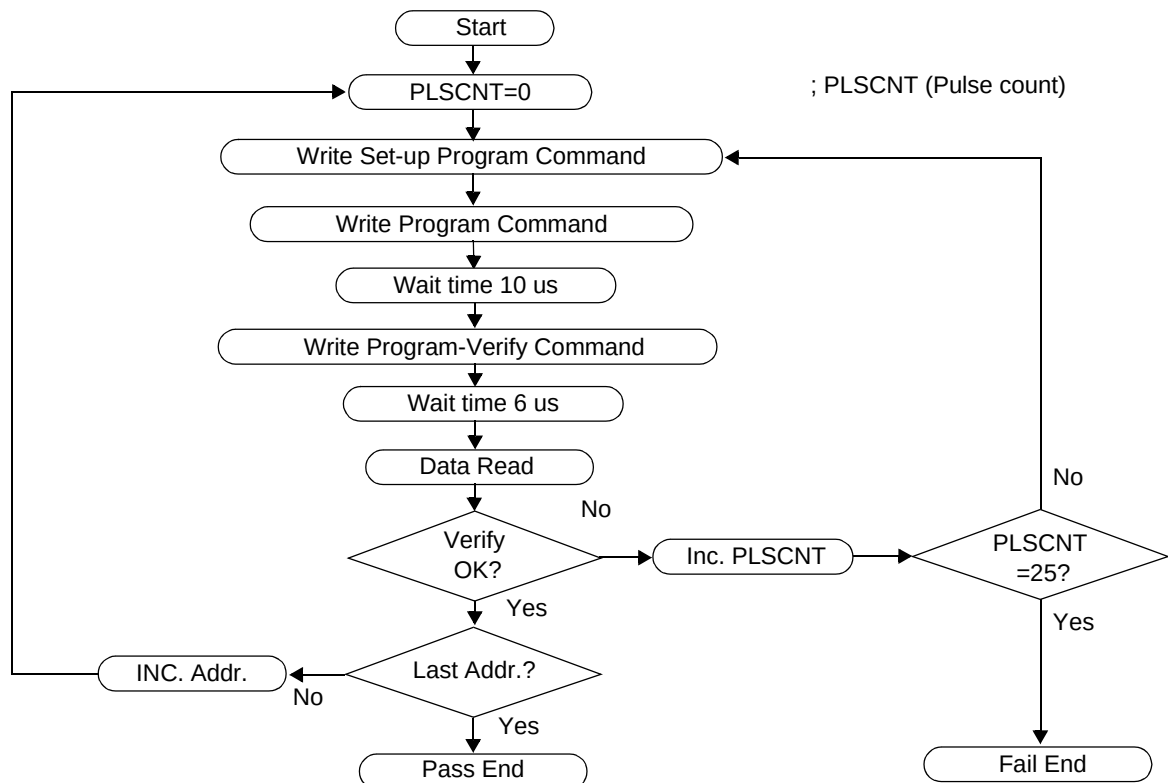


Figure 4-1 Flowchart of Standard Write Test of NOR-Type Flash Memory

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

4.1 NOR-type Flash Memory Write Test Example

4.1.1 Write Test Example - 1 (Flag Sense Instruction Offset Time: 1.4 μ s)

A sample program of a write test for parallel measurement on eight 128 KW $\times$ 8-bit flash memory chips is given below. The T5377, T5376, T5375 and T5371 can execute a write test on the condition that the offset time for the flag sense instruction is set to 1.0 μ s.

- Socket program (For T5335, T5335P and T5334)

```

SOCKET FLSOC001
  WE=PD29=29
  CE=PD30=30
  DEFINE MULTIPLE 8DUT 1STN
  BIT = 8
  PPS = 2
  PIN MODE = B
  CHTYPE2

  DUT# = VS1 ,VS2 ,PD1-18 ,PD29-30 ,P33-40
  DUT1 = PPS1,PPS17,PD1-18A1,PD29-30A1,P33-40A
  DUT2 = PPS2,PPS18,PD1-18A3,PD29-30A3,P41-48A
  DUT3 = PPS3,PPS19,PD1-18A2,PD29-30A2,P97-104A
  DUT4 = PPS4,PPS20,PD1-18A4,PD29-30A4,P105-112A
  DUT5 = PPS5,PPS21,PD1-18B1,PD29-30B1,P33-40B
  DUT6 = PPS6,PPS22,PD1-18B3,PD29-30B3,P41-48B
  DUT7 = PPS7,PPS23,PD1-18B2,PD29-30B2,P97-104B
  DUT8 = PPS8,PPS24,PD1-18B4,PD29-30B4,P105-112B
END

```

- Socket program (For T5371, T5336 and T5311)

```

SOCKET FLSOC001
  WE=PD26=26
  CE=PD29=29
  DEFINE MULTIPLE 8DUT 1STN
  BIT = 16
  PPS = 2
  PIN MODE = A
  CHTYPE6

  DUT# = VS1 ,VS2 ,PD1-18 ,PD26,PD29 ,P33-48
  DUT1 = PPS1,PPS97 ,PD1-18A1,PD26A1,PD29A1,P33-48A1
  DUT2 = PPS2,PPS98 ,PD1-18A3,PD26A3,PD29A3,P33-48A3
  DUT3 = PPS3,PPS99 ,PD1-18A2,PD27A2,PD30A2,P97-112A1
  DUT4 = PPS4,PPS100,PD1-18A4,PD27A4,PD30A4,P97-112A3
  DUT5 = PPS5,PPS101,PD1-18B1,PD26B1,PD29B1,P33-48B1
  DUT6 = PPS6,PPS102,PD1-18B3,PD26B3,PD29B3,P33-48B3
  DUT7 = PPS7,PPS103,PD1-18B2,PD27B2,PD30B2,P97-112B1
  DUT8 = PPS8,PPS104,PD1-18B4,PD27B4,PD30B4,P97-112B3
END

```

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

4.1 NOR-type Flash Memory Write Test Example

- Test Plan Program

```
PRO FLPR0001 FLS0C001
```

```
VS1=5V ;VCC
VS2=12V,R(16V) ;VPP
```

```
IN1=5V,3V
OUT1=2.4V,0.8V
```

```
P1-8 =IN1,NRZA,ACLK1,<X0-7> ;ADDRESS
P9-17 =IN1,NRZA,ACLK1,<Y0-8> ;ADDRESS
P18 =IN1,/RZX,BCLK1,CCLK1 ;/OE
WE =IN1,/RZX,BCLK2,CCLK2 ;/WE
CE =IN1,/RZX,BCLK3,CCLK3 ;/CE
P33-40=IN1,NRZA,ACLK2,I0C,OUT1,STRB1,MATCH,<D0-7>;DATA
```

	STANDBY	PROGRAM SETUP	PROGRAM TS3	PROGRAM VERIFY TS4	PROGRAM DATA-READ TS5
RATE=	150NS	100NS	10US	6US	1.43US
ACLK1=	0NS	0NS	0NS	0NS	0NS ;ADDRESS
ACLK2=	0NS	0NS	0NS	0NS	0NS ;DATA
BCLK1=	0NS	OPEN	OPEN	OPEN	0NS ;/OE(RESET)
CCLK1=	OPEN	0NS	OPEN	OPEN	100NS ;/OE(SET)
BCLK2=	OPEN	10NS	10NS	10NS	OPEN ;/WE(RESET)
CCLK2=	OPEN	55NS	55NS	55NS	OPEN ;/WE(SET)
BCLK3=	10NS	10NS	10NS	10NS	10NS ;/CE(RESET)
CCLK3=	55NS	55NS	55NS	55NS	55NS ;/CE(SET)
DREL1=	0NS	0NS	0NS	0NS	0NS
DRET1=	0NS	75NS	75NS	75NS	0NS
STRB1=	0NS	0NS	0NS	0NS	30NS

```
SELECT FLASH MODE1 H<FLAG,C0> C<FLAG,C0> TYPE1
SET FLASH LIMIT ALL 25
```

```
RESET AFM ALL
AFM MULTIMODE = ALL
AFM ADDRESS = X0-7,Y0-8
AFM BITSIZE = 8
AFM ACTION = AF(C1)>FMX,BF(C1)>FMX
```

```
REG MPAT PC=#0
MEAS MPAT MFLASH
```

```
LOGI[1] COUNT(32768)
READ AFM AF (X0-255,Y0-127) TO COUNT
```

```
STOP
```

```
END
```

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

4.1 NOR-type Flash Memory Write Test Example

- Pattern program (For T5377, T5376, T5375, T5371, T5336, T5311, T5335, T5335P and T5334)

```
MPAT FLMPA001
```

```
REGISTER
```

```
  IDX1=#1FFFE
```

```
  IDX8=#17
```

```
  XMAX=#FF
```

```
  YMAX=#1FF
```

```
  TPH=#55
```

```
  BAR=ST1
```

```
SDEF PSCMD=TP<#40
```

```
SDEF PVCMD=TP<#C0
```

```
SDEF RWDAT=TP<TPH
```

```
SDEF ACLR =XB<0 YB<0
```

```
SDEF AINC =XB<XB+1 YB<YB+1 ^ BX
```

```
START #0
```

```
  NOP          ACLR PSCMD
```

```
ST0: NOP          RWDAT TS2 ;WRITE PROGRAM SET-UP
```

```
  NOP          PVCMD TS3 ;WRITE PROGRAM & WAIT TIME 10US
```

```
  NOP          RWDAT TS4 ;WRITE VERIFY CMD. & WAIT TIME 6US
```

```
  FLGLI1 ST0    PSCMD TS5 ;VERIFY
```

```
ST1: NOP C1
```

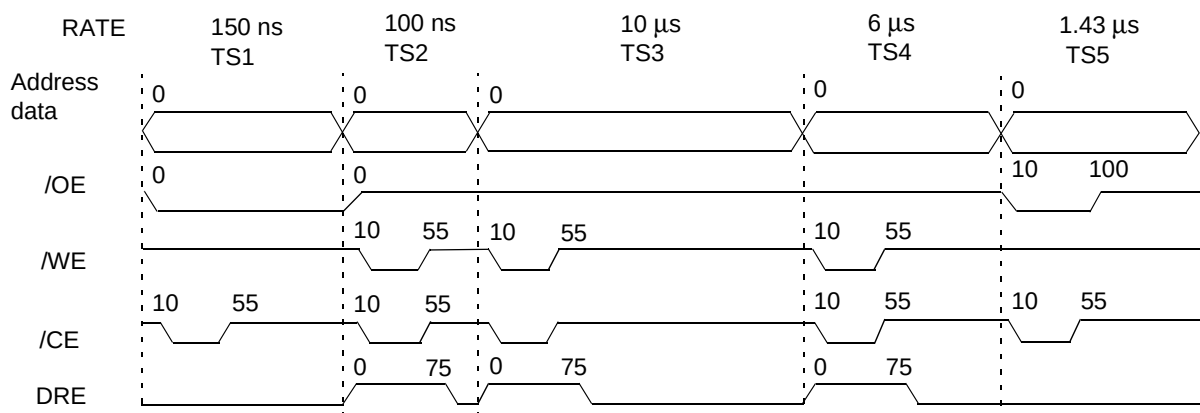
```
  JNLI1 ST0 AINC C0
```

```
  STPS
```

```
END
```

--- (10)

- Timing chart



T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

4.1 NOR-type Flash Memory Write Test Example

- Program description

- (1) Specify the socket program for the test system you are using as shown below.

For the T5335, T5335P and T5334

"BIT=8" : Specify x8-bit I/O.

"PIN MODE=A" : Specify the pin configuration for parallel measurement.

"CHTYPE2" : Specify the 650/1300CH for the TH type.

"PPS=2" : Specify the number of DUT power supplies to be used for one DUT.

For the T5371, T5336 and T5311

"BIT=16" : Specify x16-bit I/O.

"BIT=16" is specified because of insufficient driver pins for the "BIT=8" setting.

"PIN MODE=A" : Specify the pin configuration for parallel measurement.

"CHTYPE6" : Specify the 1664CH for the TH type.

"PPS=2" : Specify the number of DUT power supplies to be used for one DUT.

- (2) Allocates the device power supplies and tester pins to DUTs for parallel measurement.
- (3) Specify the voltage setting for a DUT power supply.
Specify "VS1" for the DUT VCC.
Specify "VS2" for the VPP to be used when a program is executed.
"16V PPS" is used as the VPP.
- (4) Sets the driver pin source signal and comparator pin compare conditions for device tests.
Specify "MATCH" for flash memory write tests to perform rewrite and prevent excessive write using the match detecting function of the flash memory parallel measurement.
- (5) Specifies the clock edge timings for device testing. See the timing chart for the timings programmed by this sample program.
- (6) Sets the flash memory parallel measurement functions.
"MODE1" specifies controlling the flash memory parallel measurement functions using the comparator pin match detecting function.
"H<FLAG,C0>" specifies using the hold register load conditions in the cycle for which a flag sense instruction has been specified and the clear conditions in the cycle for which "C0" has been specified.
"C<FLAG,C0>" also specifies the flash counter count and clear conditions, like that for the hold register.
"TYPE1" specifies the driver pins to be prevented against excessive write. In this example, "/WE" driver pins are specified.
"SET FLASH LIMIT ALL 25" sets limit count 25 for pass/fail decision of all devices by the write count at each address.

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

4.1 NOR-type Flash Memory Write Test Example

- (7) Sets the input conditions for the address fail memory.

In this example, the lower 8-bits of counter value data is stored in the AFM AF in the cycle when "C1" is specified. The hold register data is stored in the AFM BF in the cycle when "C1" is specified. (This function cannot be used for the T5336 and T5311.)

- (8) Starts the flash memory write test.

- (9) Read the write count for each address after the write test from AFM. (This function cannot be used for the T5336 and T5311.)

If the counter value is #3, the readout data is as follows.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
LOGI[1]	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0

- (10) Pattern program

Sets in the IDX1 register the address length to be tested - 2 times.

Sets in the IDX2 register the rewrite count - 2 times. This program executes "FLGLI1" for the IDX8 register set value + 2 times, and then jumps to the address indicated by the BAR register if the match detecting result is UNMATCH. Test pattern generation begins at address #0. After flash memory write operation, the match detecting function of the "FLGLI1" instruction is used to check whether the write operation has been correctly carried out. If writing is not correctly carried out for at least one device, rewrite processing starts. At this time, the flash memory parallel measurement starts according to the hold register load condition and counter count condition "FLAG" specified in the flash mode. For devices on which writing has been correctly carried out, driver output and logical compare are inhibited. For those on which writing has not been correctly performed, the write operations are counted.

Processing goes to the next address, if writing is not completed after rewriting for the IDX8 register set value + 2 times or if writing has completed for all devices.

In this case, the hold register and counter are cleared by "C0" specification; thus, inhibits of the driver output and logical compare are reset and the write count is cleared.

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

4.1 NOR-type Flash Memory Write Test Example

4.1.2 Write Test Example - 2 (Setting Flag Sense Instruction Offset Time 0.7 us for Rewriting)

A flag sense instruction is used in a flash memory write test to determine whether to rewrite or not. Because this instruction requires an offset time (usually 1.4 us), the testing time increases in proportion to the rewrite count. Therefore, to reduce the testing time, the flag sense offset time for rewriting should be reduced to 0.7 us. This function must satisfy the following conditions:

- (1) In the flash memory write loop (flag sense instruction loop), address/data operation instructions cannot be described. Instructions setting fixed data such as "TP<#40" are available.
- (2) The rate of the flag sense instruction (HFLGLI) must satisfy the following:

$$\text{STRB1 set value} + 300 \text{ ns} \geq \text{RATE} \times 2$$

$$0 \text{ ns} \leq \text{Each timing edge set value} \leq \text{RATE} \times 2 - 62.5 \text{ ps}$$

A sample pattern program is given below. The socket and test plan programs should be changed so that the timing settings for the flag sense instruction satisfy the conditions given above (same as Test Example - 1).

- Pattern program (For T5335, T5335P, T5334, T5336 and T5311)

```
MPAT FLMPA002
```

```
REGISTER
```

```
  IDX1=#1FFFE
```

```
  IDX8=#17
```

```
  XMAX=#FF
```

```
  YMAX=#1FF
```

```
  TPH=#55
```

```
  BAR=ST1
```

```
SDEF PSCMD=TP<#40
```

```
SDEF PVCMD=TP<#C0
```

```
SDEF RWDAT=TP<TPH
```

```
SDEF ACLR =XB<0 YB<0
```

```
SDEF AINC =XB<XB+1 YB<YB+1 ^ BX
```

Flag sense instruction loop

```
START #0
```

```
  NOP          ACLR PSCMD
```

```
ST0: NOP          RWDAT TS2 ;WRITE PROGRAM SET-UP
```

```
  NOP          PVCMD TS3 ;WRITE PROGRAM & WAIT TIME 10US
```

```
  NOP          RWDAT TS4 ;WRITE VERIFY CMD. & WAIT TIME 6US
```

```
  HFLGLI ST0 PSCMD TS5 ;VERIFY
```

```
ST1: JN11 ST0 AINC C0
```

```
  STPS
```

```
END
```

4.1.3 Write Test Example - 3 (Setting Flag Sense Instruction Offset Time 0.15 us for Rewriting)

The flag sense instruction offset time for rewriting can be reduced to 0.15 us, from that of Test Example - 2 above, by another function. This function allows to determine the match detecting result and control the flash memory parallel measurement in the flag sense instruction cycle. It changes the pattern generation sequence by the match detecting result after the fixed cycles + fixed time has passed. Control jumps to the address indicated by the ISP register when writing to all device has completed or when the flag sense instruction has been executed for the specified number of times.

At this time, the patterns generated within the fixed cycles + fixed time are given to the device as dummy patterns. For the write operation using the dummy patterns, excessive writing is prevented by flash memory parallel measurement function.

This function must satisfy the following conditions:

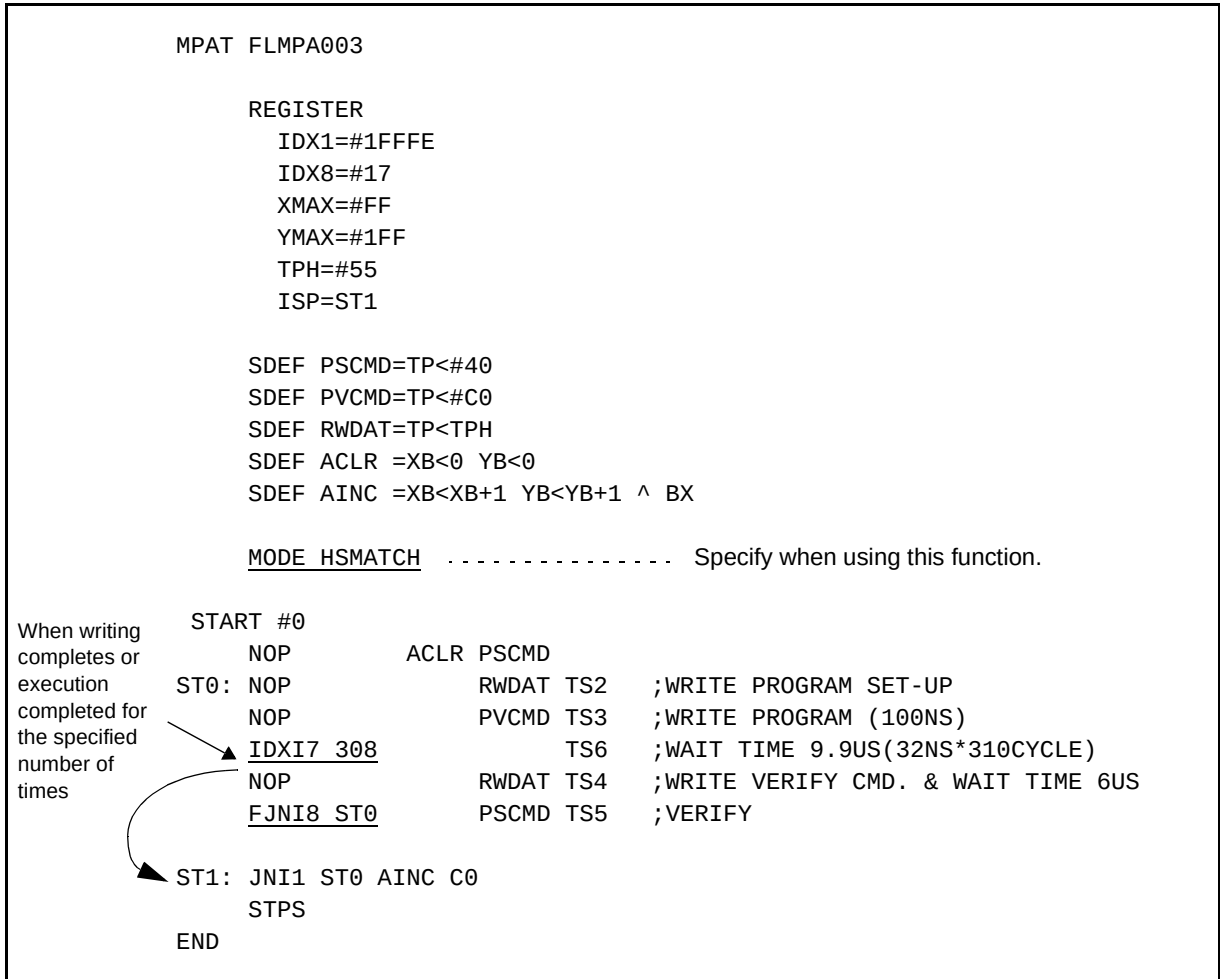
- (1) In the flash memory write loop (flag sense instruction loop), address/data operation instructions cannot be described. Instructions setting fixed data such as "TP<#40" are available.
- (2) The rate of the flag sense instruction must be set so that it satisfies "STRB1 set value + 150 ns ≤ RATE".
- (3) The pattern program must be written so that the number of patterns and time in the flag sense instruction loop exceeds "23 cycles + 700 ns + STRB set value".
- (4) Flag sense instruction execution count ≥ Limit register set value

A sample pattern program is given below. The socket and main programs should be changed so that the timing settings for the flag sense instruction satisfy the conditions given above same as "4.1.1 Write Test Example - 1 (Flag Sense Instruction Offset Time: 1.4 ms)".

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

4.1 NOR-type Flash Memory Write Test Example

- Pattern program (For T5335, T5335P, T5334, T5336 and T5311)

**CAUTION!**

When **MODE HSMATCH** is specified, the index command that can be used in the loop of the **FJNI8** command is restricted to **IDXI7** and **JNI7**.

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

4.2 NOR-type Flash Memory Erase Test Example

4.2 NOR-type Flash Memory Erase Test Example

In the erase test, each address is verified after erasing the entire chip or block; if it is not erased, re-erasing is carried out by incrementing the erase count.

Verify after the re-erase restarts where unerased data has been detected.

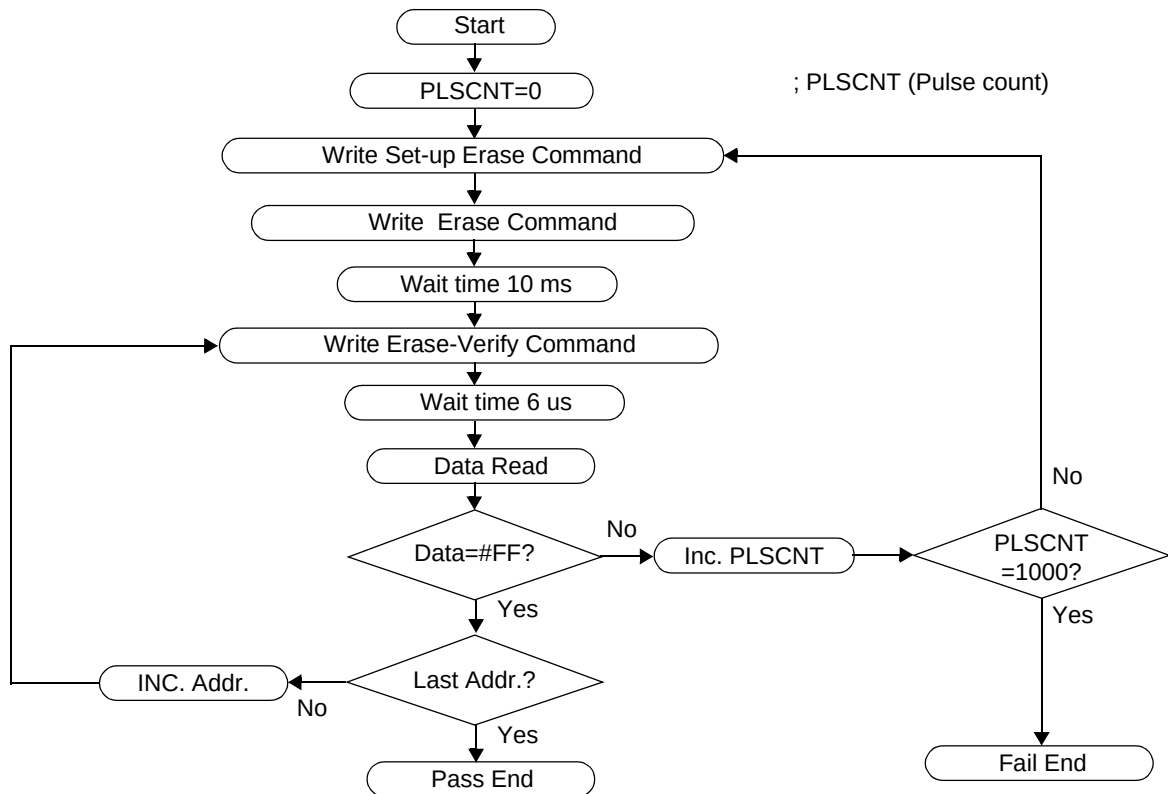


Figure 4-2 The typical erase test flowchart for NOR-type flash memory

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

4.2 NOR-type Flash Memory Erase Test Example

A sample program of an erase test for parallel measurement on eight 128KW × 8-bit flash memory chips is given below. The socket program is the same as that for FLSOC001.

- Test plan program

```

PRO FLPR0002 FLSOC001
  VS1=5V ;VCC
  VS2=12V, R(16V) ;VPP
  IN1=5V, 3V
  OUT1=2.4V, 0.8V

  P1-8 =IN1, NRZA, ACLK1, <X0-7> ;ADDRESS
  P9-17 =IN1, NRZA, ACLK1, <Y0-8> ;ADDRESS
  P18 =IN1, /RZX, BCLK1, CCLK1 ;/OE
  WE =IN1, /RZX, BCLK2, CCLK2 ;/WE
  DE =IN1, /RZX, BCLK3, CCLK3 ;/CE
  P33-40=IN1, NRZA, ACLK2, IOC, OUT1, STRB1, MATCH, <D0-7>;DATA

;          ERASE   ERASE   ERASE
;          STANDBY SETUP   VERIFY   DATA-READ
;          TS1      TS2     TS3      TS4      TS5
RATE= 150NS 100NS 10MS 6US 1.43US
ACLK1= 0NS 0NS 0NS 0NS 0NS ;ADDRESS
ACLK2= 0NS 0NS 0NS 0NS 0NS ;DATA
BCLK1= 0NS OPEN OPEN OPEN 0NS ;/OE(RESET)
CCLK1= OPEN 0NS OPEN OPEN 100NS ;/OE(SET)
BCLK2= OPEN 10NS 10NS 10NS OPEN ;/WE(RESET)
CCLK2= OPEN 55NS 55NS 55NS OPEN ;/WE(SET)
BCLK3= 10NS 10NS 10NS 10NS 10NS ;/CE(RESET)
CCLK3= 55NS 55NS 55NS 55NS 55NS ;/CE(SET)
DREL1= 0NS 0NS 0NS 0NS 0NS
DRET1= 0NS 75NS 75NS 75NS 0NS
STRB1= 0NS 0NS 0NS 0NS 30NS

SELECT FLASH MODE1 H<FLAG,C0> C<FLAG, FIXL> TYPE2
SET FLASH LIMIT ALL 1000

REG MPAT PC=#0
MEAS MPAT FLMPA004.

STOP
END

```

(1)

(2)

(3)

(4)

(5)

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

4.2 NOR-type Flash Memory Erase Test Example

- Pattern program

```

MPAT FLMPA004

REGISTER
  IDX1=#1FFFE
  XMAX=#FF
  YMAX=#1FF
  TPH=#FF

SDEF ERCMD=TP<#20
SDEF EVCMD=TP<#A0
SDEF RWDAT=TP<TPH
SDEF ACLR =XB<0 YB<0
SDEF AINC =XB<XB+1 YB<YB+1 ^ BX

START #0
  NOP      ACLR      ERCMD
ST0: NOP      TS2      ;WRITE ERASE SET-UP
  NOP      C0 EVCMD TS3 ;WRITE ERASE      & WAIT TIME 10MS
ST1: NOP      RWDAT TS4 ;WRITE VERIFY CMD. & WAIT TIME 6US
  FLGL1 ST0      ERCMD TS5 ;VERIFY
  JN11 ST1 AINC C0 EVCMD
  STPS
END
  
```

(6)

- Program description

(1) Sets the device power supply voltage and the device I/O voltage.

(2) Sets the pin conditions for device tests.

For flash memory erase tests, specify "MATCH" for comparator pins to be put to determine whether to perform re-erasing and prevent excessive erase using the match detecting function of the flash memory parallel measurement.

(3) Specifies the clock edge timings for erase tests.

(4) Sets the flash memory parallel measurement functions.

"MODE1" specifies controlling the flash memory parallel measurement functions using the comparator pin match detecting function.

"H<FLAG,C0>" specifies using the hold register load conditions in the cycle for which a flag sense instruction has been specified and the clear conditions in the cycle for which "C0" has been specified.

"C<FLAG,FXL>" specifies the flash counter count and clear conditions. If "FXL" is specified as the clear condition, clear is inhibited over the entire cycle.

"TYPE2" specifies the driver pins to be prevented against excessive erase. In this example, "/CE" driver pins are specified.

"SET FLASH LIMIT ALL1000" sets limit count 1000 for pass/fail decision of all devices by the erase count.

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

4.2 NOR-type Flash Memory Erase Test Example

(5) Starts the flash memory erase test.

(6) Pattern program

Sets in the IDX1 register the address length to be tested - 2 times.

Test pattern generation begins at address #0. After flash memory erase operation over the entire chip, the match detecting function of the "FLGL1" instruction is used to check whether the erase operation has been correctly carried out from the first address. If all devices have been erased, the hold register is cleared by "C0" description before going to the next address.

If there is at least one device not erased, re-erase processing starts.

At this time, the flash memory parallel measurement starts according to the hold register load condition and counter count condition "FLAG" specified in the flash mode. For devices where erasing has been correctly carried out, driver output and logical compare are inhibited. For those not erased, the erase operations are counted. After re-erase, the hold register is cleared by the "C0" specification and verify restarts from the address at which data not erased has been detected.

When the erase count reaches the limit register set value, the device is determined as fail.

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

4.3 Program Example for Verifying Flash Memory in Block Units

4.3 Program Example for Verifying Flash Memory in Block Units

For the NAND-type flash memory write test, data is written in block units, and then written data is verified over the entire blocks. If writing is not correctly carried out at least one address, write processing is performed again in block k units.

There is a fail stack register to store fail information of the logical compare result in such a test. A flag sense instruction is used to determine whether to carry out writing again based on the result. In addition, the result allows to control the flash memory parallel measurement function, prevent excessive write, and decide pass/fail based on the write count.

A sample program for verifying in block units is given below.

The socket program is used for this program as the same as the one used in Section 4.1.1, “Write Test Example - 1 (Flag Sense Instruction Offset Time: 1.4 ms).”

```

PRO FLPR0003 FLS0C001
  SELECT FLASH MODE2 H<FLAG,C0> C<FLAG,C0> TYPE1      (1)
  SET FLASH LIMIT ALL 20

  REG MPAT PC=#0
  MEAS MPAT FLMPA005 ..... (2)

END

```

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

4.3 Program Example for Verifying Flash Memory in Block Units

```
MPAT FLMPA005
```

```
REGISTER
```

```
  IDX1=#FE
```

```
  IDX2=#FFE
```

```
  XMAX=#FF
```

```
  YMAX=#FFF
```

```
  TPH=#55
```

```
  BAR=ST2
```

```
SDEF ACLR = XB<0 YB<0 TP<TPH
```

```
SDEF AXINC= XB<XB+1
```

```
SDEF AYINC= YB<YB+1
```

(3)

```
START #0
```

```
  NOP ACLR
```

```
ST0: JN11 ST0 AXINC W TS2 ;BLOCK PROGRAM
```

```
ST1: JN11 ST1 AXINC R TS3 ;BLOCK VERIFY
```

```
  NOP TS4 ;DUMMY
```

```
  NOP TS4 ;DUMMY
```

```
  NOP TS4 ;DUMMY
```

```
  NOP TS4 ;DUMMY
```

```
  FLGLI1 ST0 TS5 ;BLOCK MATCH CHECK
```

```
ST2: JN12 ST0 AYINC C0 TS6 ;NEXT BLOCK
```

```
STPS
```

```
END
```

- Program description

- (1) Sets the flash memory parallel measurement functions.

“MODE2” specifies controlling the flash memory parallel measurement functions using the fail stack register information.

“H<FLAG,C0>” specifies using the hold register load conditions in the cycle for which a flag sense instruction has been specified and the clear conditions in the cycle for which “C0” has been specified.

“C<FLAG,FXL>” specifies the flash counter count and clear conditions.

“TYPE1” specifies the driver pins to be prevented against excessive write.

“SET FLASH LIMIT ALL 20” sets limit count 20 for pass/fail decision of all devices by the write count of each block.

- (2) Specifies starting the flash memory verify test in block units.

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

4.3 Program Example for Verifying Flash Memory in Block Units

(3) Pattern program

Data is written continuously in 256 words of a block, and then it is continuously verified in 256 words by logical compare of pass/fail. The fail information is stacked in the fail stack register and the stacked information is determined using the match detecting function of the "FLGLI1" instruction. If a fail is found, rewrite processing starts.

Matching detection using the fail stack register data requires the following number of dummy cycles after the end of comparison cycle.

System name	Number of dummy cycles
T5377 T5376 T5375	10
T5371	5
T5336 T5311 T5335 T5335P T5334	4

At this time, the flash memory parallel measurement function starts according to the hold register load conditions and counter count conditions "FLAG" specified in the flash mode. For passed devices, the driver output and logical compare are inhibited. For failed devices, the write operations are counted. The fail stack register is cleared by the hold register load conditions.

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

4.4 Fast Erase Test on Flash Memory Using MEAS MPAT(5)

4.4 Fast Erase Test on Flash Memory Using MEAS MPAT(5)

In the flash memory parallel test using the standard erase test sequence as shown in Section 4.2, the re-erase operation is carried out for only those devices in which a verification fail has occurred. At that time, the erase operation is inhibited for a passed device to prevent excessive erase. This lowers the possibility of having to erase all devices in parallel, but increases the erase operation count in parallel measurement, thus increasing the testing time.

To solve this problem, the fast erase test re-erases all devices after detecting verification fails in all devices. Verification after the erase operation is re-started from the address where the first fail occurred in all devices. Thus the erase operation count is reduced and testing time is shortened.

The following shows the operation images of the fast erase test.

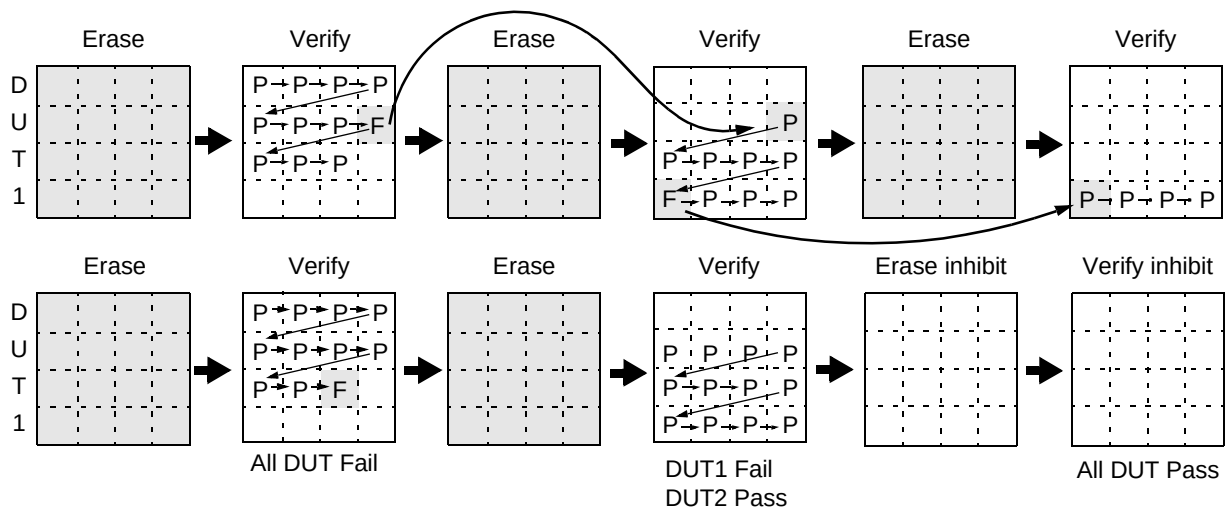


Figure 4-3 Fast erase test operation images

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

4.4 Fast Erase Test on Flash Memory Using MEAS MPAT(5)

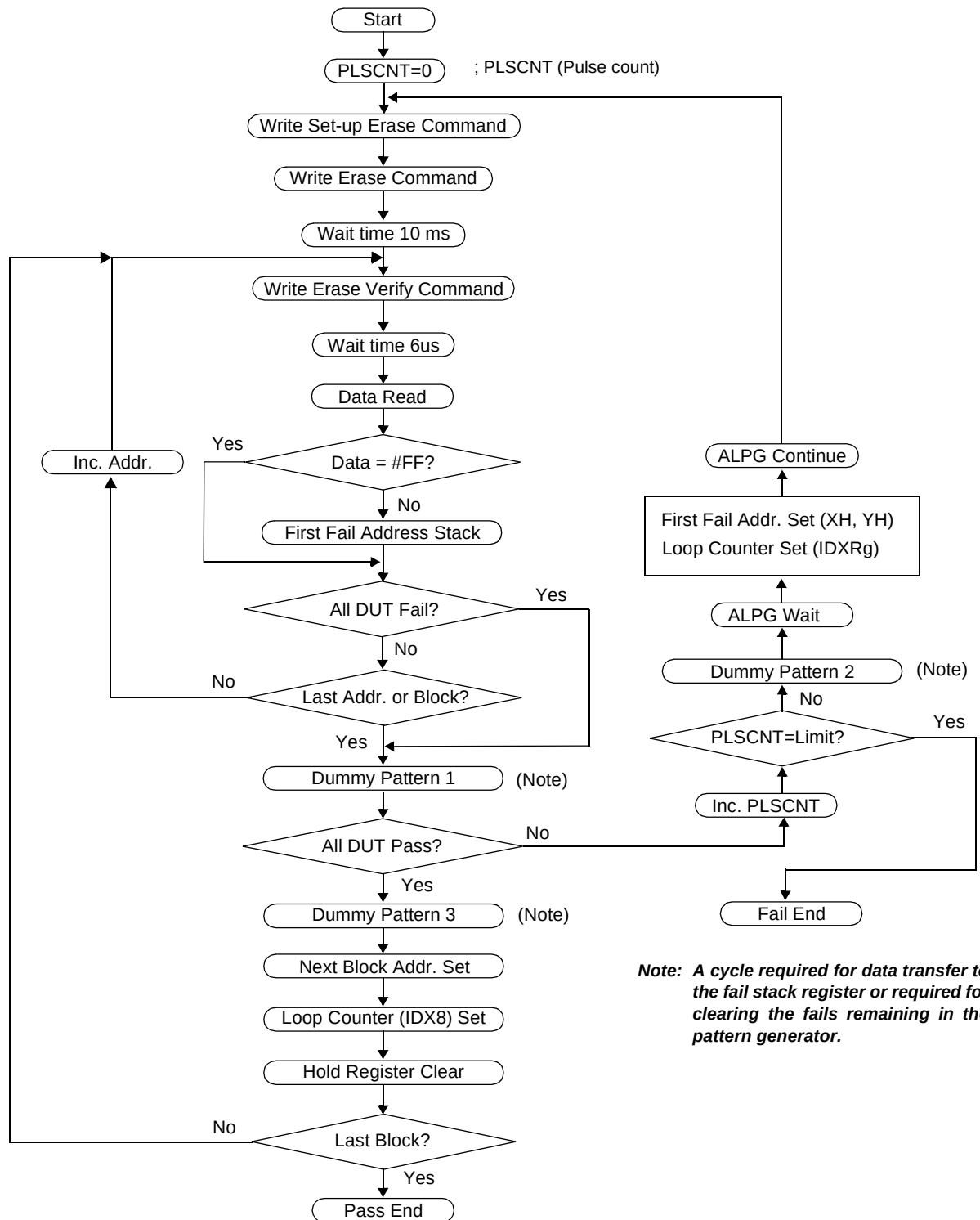


Figure 4-4 Fast erase test method flowchart

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

4.4 Fast Erase Test on Flash Memory Using MEAS MPAT(5)

4.4.1 Block Division

The fast erase test algorithm is a method for minimizing the erase operation execution count in the parallel measurement. This method is not efficient because the verification execution count after the erase operation increases depending on the distribution of the erase count for each address of the respective devices in parallel measurement.

For example, in the worst case, when fails are distributed in the first address of DUT1 and the last address of DUT2, all addresses of each device are subjected to verification, so that the time for verification becomes longer than that for erasing (see the example below). To solve this problem, it is necessary to further divide the area to be erased in the device itself into blocks and execute the test by specifying the area to be verified, thus reducing the verification time.

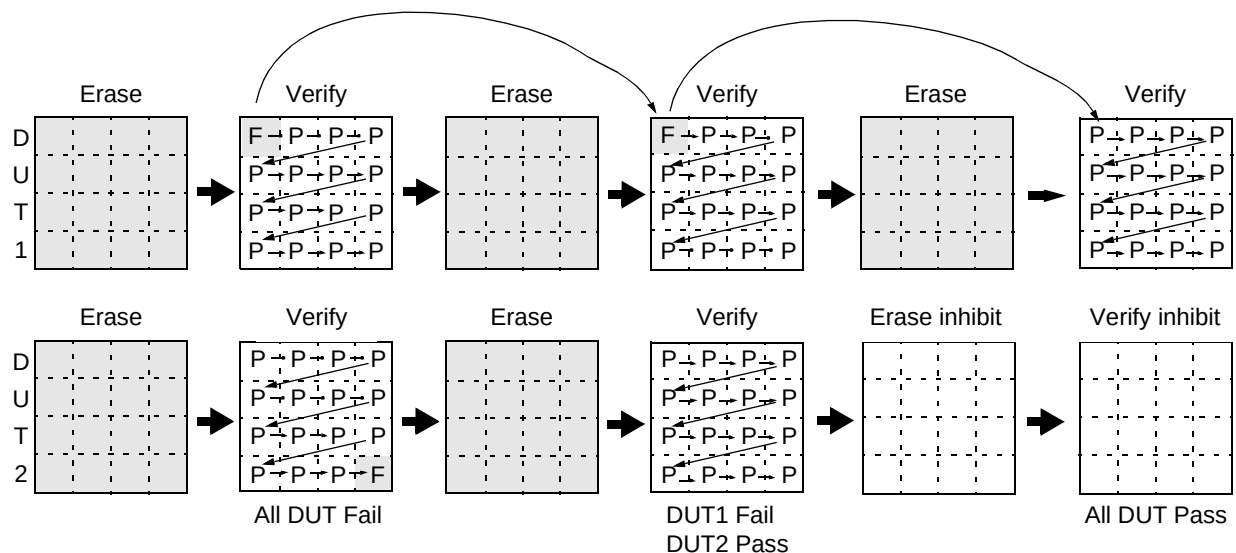


Figure 4-5 Verify Operation for Fails Distributed at the First Address of DUT1 and the Last Address of DUT2

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

4.4 Fast Erase Test on Flash Memory Using MEAS MPAT(5)

4.4.2 Calculating the Optimum Block Division

The effect of the fast erase test algorithm differs according to the size of a divided block. Thus, when this test algorithm is applied, it is necessary to calculate the most effective size of a divided block for each type of measured device through actual measurement.

$$\text{Erase test time} = (\text{Single erase time}) \times (\text{Erase count}) + (\text{Single verification time}) \times (\text{Verification count})$$

Erase count: Decreases as the area to be verified is larger.

Verification count: Increases as the area to be verified is larger.

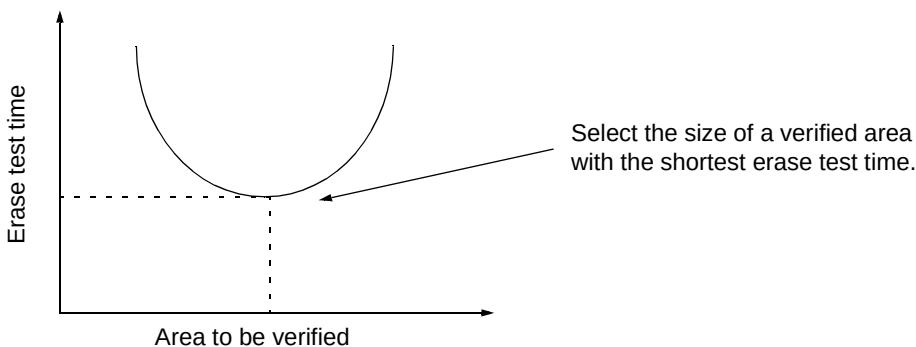


Figure 4-6 Relationship Between Erase Time and Verified Area

The following two examples show the fast erase test algorithm as applied to dividing the 512 K-word area (X: #0 to #3FF, Y: #0 to #1FF).

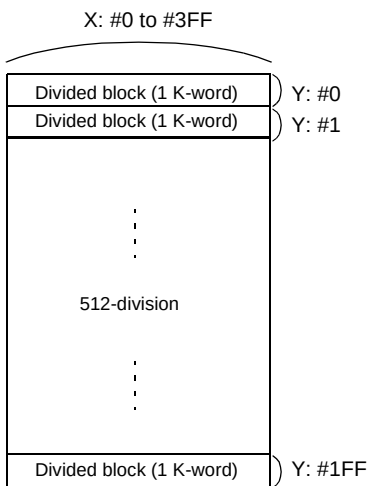


Figure 4-7 Dividing a 512K-Word Area into 1K-Word Areas

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4.4 Fast Erase Test on Flash Memory Using MEAS MPAT(5)

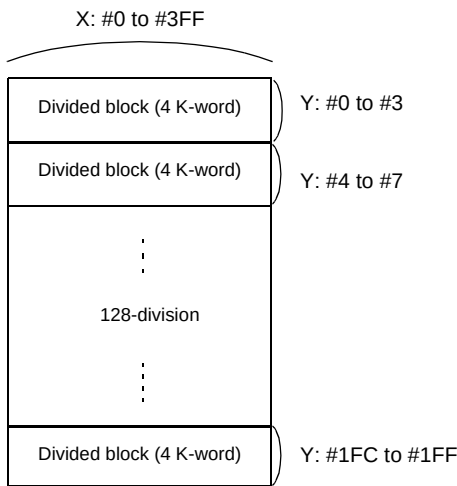


Figure 4-8 Dividing a 512K-Word Area into 4K-Word Areas

4.4.3 Sample Program 1 for the Fast Erase Test Method (Chip Erase)

The following shows a sample program for chip erase using the fast erase test algorithm. This sample program uses a 1 K-word block division size (See Figure 4-7). The socket program is the same as for the “4.1.1 Write Test Example - 1 (Flag Sense Instruction Offset Time: 1.4 ms)”.

- Test plan program

```

PRO FLPR0003 FLSOC001
.....
AFM MULTIMODE = ALL
SELECT FLASH MODE2 H<FLAG,C3> C<FLAG, FIXL> TYPE2 ----- (1)
SET FLASH LIMIT ALL 1000
REG MPAT PC = #0
MEAS MPAT(5) FLMPA006 ----- (2)
STOP
END
    
```

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

4.4 Fast Erase Test on Flash Memory Using MEAS MPAT(5)

- Pattern program

```

MPAT FLMPA006
SDEF INIT = XB<XH YB<YH XCS<0 YCS<0 TP<TPH
SDEF HT0B = XB<XH YB<YH
SDEF BT0C = XC<XB YC<YB
SDEF ST0CS= XCS<XS YCS<YS+D1
SDEF C0UT = X<XC Y<YC
SDEF CINC = XC<XC+1 YC<YC+1^CY
SDEF HC = C3 ;Hold Register Clear

REGISTER
XMAX = #3FF ;X address size - 1
YMAX = #1FF ;Y address size - 1
IDX2 = #1FE ;Divided block size - 2 ----- (3)
IDX8 = #3FE ;Divided block size - 2 (Set the same value for STI8.)
XH = #0 ;Verification start address
YH = #0
D1 = #1 ;Y address size of divided block ----- (3)
TPH = #FF
DRE1 = W
CPE1 = R

START #00
NOP INIT TS1 ;
L0: NOP C0UT BT0C TS1 ; ----- (4)
NOP C0UT TP<#20 TS1 ;
NOP C0UT TP<#20 TS2 C0 W ;WRITE SET-UP ERASE COMMAND
NOP C0UT TS3 C0 W ;WRITE ERASE COMMAND & WAIT TIME 10 MS
L1: NOP C0UT TP<#A0 TS1 ;
JAF L2 C0UT TP<#FF TS4 C0 W ;WRITE SET-UP VERIFY COMMAND
JNI8 L1 C0UT CINC TS5 C0 C1 R ;WAIT TIME 6 US & DATA READ } ----- (5)

L2: IDXI1 #6 C0UT TS1 ;DUMMY 1
FLGS1 L3 C0UT TS6 ;MATCH ----- (6)
IDXI1 #14 C0UT TS1 ;DUMMY 2
WAIT ;RESTORE VERIFY START ADDRESS ----- (7)
JMP L0 C0UT HT0B TS1 ; ----- (8)

L3: IDXI1 #14 C0UT TS1 ;DUMMY 3
STI8 #3FE C0UT ST0CS TS1 HC ;NEXT BLOCK ADDR. (YCS<YS+D1) ----- (9)
JNI2 L1 C0UT TS1 ;
STPS
END

```

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

4.4 Fast Erase Test on Flash Memory Using MEAS MPAT(5)

- Program description

- (1) Sets the flash memory parallel measurement functions.

When "MODE2" is specified, device output data is logically compared for the pass/fail result, the fail information is stored in the fail stack register, and match detection is executed with this fail information.

"H<FLAG,C3>" specifies the cycle with the flag sense instruction for the loading conditions to the hold register and the cycle with MUT command C3 for the clear conditions.

"C<FLAG,FXL>" specifies the cycle with the flag sense instruction for the flash counter count conditions and clear inhibition for the clear conditions.

"TYPE2" specifies output inhibition for the "/CE" driver pin to prevent excessive erase.

"SET FLASH LIMIT ALL 1000" specifies an erase count limit of 1000 for the pass/fail determination for all devices.

- (2) Specifies starting the flash memory fast erase test by MEAS MPAT(5).

- (3) IDX2 and D1 can be obtained from the following equations.

Device size = (X address size) × (Y address size)

IDX2 = (Device size) ÷ (Divided block size) - 2

D1 = (Divided block size) ÷ (X address size)

- (4) The verification restart address (base register) is loaded to the current register.

- (5) When "MODE2" is specified for the flash memory parallel measurement functions, the address of the first fail of all the devices is stored in the fail address stack register. The pass/fail result information (fail result) of erase verification for each device is stored in the fail stack register. Based on this information, fail occurrence is monitored for all devices by the JAF instruction. When the JAF instruction indicates that all devices have a fail, the processing jumps to Label L2.

- (6) When a fail is detected in all devices or the erase verification is completed up to the final address of a divided block, the flag sense instruction determines whether to execute re-erase with the match detection function based on the fail stack register information. (When a fail is detected in all devices or a match fail is detected after the erase verification is completed up to the final address of a divided block, re-erase is executed. When a match pass is detected after the erase verification is completed up to the final address of a divided block, re-erase is not executed.)

When re-erase is executed, passed devices are subjected to logical comparison inhibition and driver inhibition according to the hold register load conditions, so that the passed devices are excluded from the test target. (The devices excluded from the test target return to become test targets according to the clear conditions.) For the failed devices, the number of erase executions is counted according to the count conditions. When the erase execution count reaches the limit value 1000, a device fail is determined, so that those devices are excluded from the test target.

When re-erase is not executed, that is, erase verification pass is detected up to the final address of a divided block, the processing jumps to Label L3 and proceeds to the next divided block.

The fail stack register information is cleared in the cycle with FLGS1 of the hold register load conditions.

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

4.4 Fast Erase Test on Flash Memory Using MEAS MPAT(5)

- (7) The address (first fail address) stored in the fail address stack register is transferred to XH and YH, and is used to calculate the remaining verification execution count (IDX8R). This first fail address is used as the verification restart address after re-erase.
- (8) The verification restart address is set (in the base register) to restart pattern generation.
- (9) The start address of the next block is set (in the current register) to reset the block size (IDX8 value). The passed devices excluded from the test target in (6) are returned to become test targets by C3 of the hold register clear conditions.

Note1: The following table lists the required number of dummy cycles for each setting of DUMMY1 to DUMMY3 in the pattern program:

System name	DUMMY1	DUMMY2	DUMMY3
T5377 T5376 T5375	13 cycles or more	70 cycles or more	70 cycles or more
T5371	10 cycles or more	40 cycles or more	40 cycles or more
T5336 T5311 T5335 T5335P T5334	8 cycles or more	22 cycles or more	22 cycles or more

Note2: For the T5377, T5376, T5375, T5371, T5336, T5311, T5335, T5335P and T5334, executing / MTRACE could not correctly trace the pattern program that includes the JAF instruction.

Note3: For the T5311 and T5336, the JAF instruction must be used with AFM MULTIMODE=ALL. A setting of AFM MULTIMODE=SINGLE causes the JAF instruction to store the failure addresses of the DUT specified in the AFM DUTSEL statement. The default setting is AFM MULTIMODE=ALL.

Note4: For the T5377, T5376, T5375 and T5371, fails cannot be written into the AFM correctly with the setting other than AFM MULTIMODE = ALL. The default setting is AFM MULTIMODE = SINGLE.

Note5: When the JAF instruction is used with the T5377, T5376 and T5375, set a test rate (RATE) 14 ns or longer for all cycles.

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

4.4 Fast Erase Test on Flash Memory Using MEAS MPAT(5)

When the divided block size is changed, the respective registers and STI8 in the index reference instruction have the following values.

Register	Divided Block Size			
	1 K	2 K	4 K	8 K
XMAX	#3FF	#3FF	#3FF	#3FF
YMAX	#1FF	#1FF	#1FF	#1FF
IDX2	#1FE	#FE	#7E	#3E
D1	#1	#2	#4	#8
IDX8	#3FE	#7FE	#FFE	#1FFE
STI8	#3FE	#7FE	#FFE	#1FFE

Each value is calculated as follows:

XMAX = X address register size - 1

YMAX = Y address register size - 1

IDX2 = Number of divided blocks - 2

= (Device size) ÷ (Divided block size) - 2

= ((X address size) × (Y address size)) ÷ (Divided block size) - 2

D1 = Y address size of the divided block

= (Divided block size) ÷ (X address size)

IDX8 = Divided block size - 2

STI8 = IDX8 (Set the same value as IDX8.)

For STI8 in the index reference instruction, set the same value as IDX8.

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

4.4 Fast Erase Test on Flash Memory Using MEAS MPAT(5)

4.4.4 Sample Program 2 for the Fast Erase Test Method (Sector Erase)

The following shows a sample program for sector erase using the fast erase test algorithm.

The socket program is used for this program is the same as the one used in Section 4.1.1, "Write Test Example - 1 (Flag Sense Instruction Offset Time: 1.4 ms)."

- Test plan program

```

PRO FLPR0004 FLSOC001

LOGICAL SECXAD(32)=#0!16,#200!16 ; Sector X-Addr.
LOGICAL SECYAD(32)=#000,#020,#040,#060,#080,#0A0,#0C0,#0E0,@ ; Sector Y-Addr.
      :          #100,#120,#140,#160,#180,#1A0,#1C0,#1E0,@
      :          #000,#020,#040,#060,#080,#0A0,#0C0,#0E0,@
      :          #100,#120,#140,#160,#180,#1A0,#1C0,#1E0
      :

DO LP I=1,32,1
  SELECT FLASH MODE2 H<FLAG,C3> C<FLAG,FXL> TYPE1
  SET FLASH LIMIT ALL 500
  REG MPAT PC =#0
  REG MPAT XOS=SECXAD(I) ..... (1)
  REG MPAT YOS=SECYAD(I) ..... (1)
  MEAS MPAT(5) FLMPA007
LP:
STOP
END

```

T5300 SERIES FLASH MEMORY PARALLEL MEASUREMENT FUNCTION DESCRIPTION

4.4 Fast Erase Test on Flash Memory Using MEAS MPAT(5)

- Pattern program

MPAT FLMPA007

REGISTER

```

XMAX = #1FF      ; X address size of sector = XMAX+1
YMAX = #1F       ; Y address size of sector = YMAX+1
IDX2 = #E        ; Divided block number - 2 = (sector size) ÷ (IDX8+2) - 2
IDX8 = #3FE      ; Divided block size - 2
XH = #0          ; Verification start address
YH = #0
D1 = #2          ; Y address size of divided block = (IDX8+2) ÷ (XMAX+1)
TPH = #FF
DRE1 = W
CPE1 = R

```

..... (1)

START #00

```

NOP      INIT      TS1      ;
L0: NOP   COUT BT0C   TS1      ;
NOP      COUT TP<#60 TS1      ;
NOP      COUT TP<#60 TS2 C0   W ;WRITE SET-UP ERASE COMMAND
NOP      COUT      TS3 C0   W ;WRITE ERASE COMMAND & WAIT TIME 10MS

```

⋮

The following part is the same part shown in Section 4.4.3, "Sample Program 1 for the Fast Erase Test Method (Chip Erase)."

- Program description

- (1) For sector erase, the fast erase test can be applied by setting each sector start address in the XOS and YOS registers for each sector. The remaining portion of program is the same as the program for "4.4.3 Sample Program 1 for the Fast Erase Test Method (Chip Erase)".